

Protecting PII Data in LLM's by using Encoding/ introducing noise at client end

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ABSTRACT

There has been ever growing concerns about the privacy in this world of Large models where anyone chatting with AI chatbots is skeptical about how his/her information will be leaked out or can be mis-used. This paper investigates various privacy-preserving methods for client data protection in Large Language Model (LLM) inference. There are already many privacy preserving chatbots in market but they either lack appropriate privacy preservation or are very computational heavy. We examine existing approaches including the Split-N-Denoise (SnD) framework and cryptographic techniques such as homomorphic encryption and various perturbation methods to analyze how they balance utility and privacy, with particular focus on local differential privacy (LDP) applications for text data.

This paper proposes the **CleanSplit Model (CSM)**, a novel privacy-preserving inference framework that deploys only a lightweight token representation layer on the client side. The primary objective of CSM is to protect user privacy during LLM inference without requiring modifications to existing model parameters, thereby optimizing the critical trade-off between privacy preservation and computational utility.

This paper evaluated various techniques to protect private data particularly PII (Personally Identifiable Information) like Faking PII data, Masking PII data, replacing PII data using client side LLM. Evaluation used various Metrics likes PII Leakage Rate, Re-identification Risk Score, Information Entropy, BLEU Score, ROUGE Scores (ROUGE-1, ROUGE-2, ROUGE-L).

As an outcome our approach demonstrated that depending upon the context of problem someone have to pick privacy protection solutions as an example, if we are using chatbot where an immediate answer is required Faking is most appropriate technique with PII leakage rate of 12.4% and latency of 287ms whereas if someone uses something which is Batch based client side LLM are more powerful in protecting PII data with with PII leakage rate of 5.8% and latency of 1243ms.

Finally, CSM proposed Faking as an that approach significantly reduces computational overhead for removing personally identifiable information (PII) at the client side—where sensitive data resides—while maintaining LLM performance efficiency and result quality.

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CHAPTER 1

INTRODUCTION

The growing concern over data privacy in Large Language Models (LLMs) is reflected in recent governmental and regulatory interventions, including the EU's AI Act, the US Executive Order on Safe, Secure, and Trustworthy AI, and established regulations such as **GDPR and HIPAA**. These frameworks emphasize the necessity for AI systems to incorporate robust privacy-preserving mechanisms, making the development of practical and efficient solutions for local privacy protection in LLM inference paramount.

1.1 Background of the Study

The escalating threat of data breaches is evident across multiple sectors, especially in healthcare and financial services, where sensitive personal identifiable information (PII) and monetary assets are at stake.

PII (Personally Identifiable Information) refers to any data that can directly or indirectly identify an individual. It's a foundational concept in data privacy, cybersecurity, and regulatory compliance — especially relevant when studying data breaches in financial institutions. according to the U.S. National Institute of Standards and Technology (NIST SP800-122): PII is “any information about an individual maintained by an agency, including any information that can be used to distinguish or trace an individual's identity.”

As per latest HIPPA report (Steve Alder, 2025) in June 2025, the healthcare sector witnessed a **16.67% monthly increase** in security incidents, and a **302.71% rise** in individuals affected by PHI breaches compared to the previous month. Tens of millions of individuals' PII are compromised monthly, with over **7.6 million** affected in just June 2025. Globally, cybercrime's economic impact is projected to reach **US\$10.5 trillion** by 2025, growing about **15% annually**.

Digital transformation has undoubtedly ushered in human-like, connected, and empowering digital experiences and efficiency in money management for individuals and businesses. Financial institutions are similarly under intense pressure and exposure. As per Journal article (Olumayowa Adefowope Adekoya, 2025)nearly **46%** of

financial institutions worldwide reported experiencing a data breach in the last 24 months. In 2024, about **65%** of financial organizations reported ransomware attacks—up from **64% in 2023** and dramatically higher than **34% in 2021**. The finance sector is currently the most affected industry under GDPR in the UK: from 2023 through Q1 2025, finance businesses self-reported **2,175** breaches.

However, this convenience has increased reliance on digital services. The increased reliance on digital services comes with an expanded cyber security attack surface for malicious actors . This expanded attack surface has led to a rise in cyber attacks. Consequently, the global average cost of a data breach increased by 10 % to 4.88 million dollars in 2024, according to IBM’s Data Breach Cost Report (IBM Corporation, 2025). As per report The cost per data breach for financial firms is among the highest of all industries. For example, in 2024, for large-scale breaches (50 million records or more compromised), financial firms faced average costs of **US\$375 million**.

Detection and containment times for financial firms are also significant: it took about **168 days** on average to detect a breach, and **51 days** to contain it. The volume of records compromised in the financial sector has soared. Between 2022 and 2023, records affected increased from **~29.3 million** to over **43.6 million** breached records across financial companies.

In terms of incident frequency, in the first half of 2025, financial services reported **387 data breaches** in the United States, making it one of the most breached industries.

These figures underscore that financial organizations do not merely face more attacks; the stakes are higher in terms of cost, scale, duration of exposure, and potential downstream harm (fraud, reputational damage, regulatory penalties). Together with the healthcare data, they paint a picture of sectors under acute threat — reinforcing the necessity of advanced defence mechanisms, robust detection/prevention, and rapid response strategies.

1.2 Problem Statement

The rapid adoption of large language models (LLMs) across sensitive domains such as healthcare, finance, and education has heightened concerns around data privacy, particularly regarding the protection of personally identifiable information (PII). While several privacy-preserving techniques have been proposed, existing methods continue to face critical limitations that hinder their effectiveness in real-world deployments.

Cryptographic approaches—such as Secure Multi-Party Computation (SMPC) and Homomorphic Encryption (HE)—offer strong theoretical guarantees for data privacy. However, they impose significant computational overhead, resulting in increased latency and resource consumption. This makes them unsuitable for time-sensitive applications that demand real-time or near-real-time inference, such as clinical decision support systems or financial transaction monitoring.

Alternatively, perturbation-based methods, including those relying on differential privacy, attempt to obfuscate sensitive information by adding statistical noise. While these methods can improve scalability, they often struggle to maintain an optimal balance between data utility and privacy. The trade-off frequently leads to diminished model performance, reduced accuracy, and unreliable outputs—especially detrimental in high-stakes environments where precision is critical.

Recently, the Split-N-Denoise (SnD) framework has emerged as a hybrid solution that introduces differential privacy at the architectural level. While it offers improvements in utility preservation compared to pure perturbation methods, SnD is not without its flaws. Notably, the injection of noise before transmission to LLM servers can still result in indirect PII leakage, especially if the noise does not fully obfuscate identifiable tokens. Moreover, the absence of integrated Named Entity Recognition (NER) mechanisms means that many identifiable entities—such as names, locations, and financial identifiers—can pass through unrecognized and unprotected. This poses a significant privacy risk, particularly in sectors governed by stringent regulations such as HIPAA, GDPR, and financial compliance frameworks.

Given these limitations, there is a pressing need for a more robust and practical solution that ensures strong privacy guarantees without sacrificing model performance or scalability. This research aims to address this gap by proposing the **CleanSplit Model (CSM)**—a novel client-side privacy-preserving inference framework. CSM incorporates a NER-driven approach to detect and replace sensitive entities prior to model interaction, ensuring that no raw PII is ever transmitted to LLM servers. Unlike cryptographic or noise-based techniques, CSM seeks to maintain computational efficiency while achieving a higher level of privacy assurance.

By focusing on real-world applications in healthcare, finance, and education—where the protection of PII is not only a legal requirement but a foundational element of user trust—this study contributes a practical, efficient, and secure framework for private LLM inference. The proposed solution is intended to bridge the gap between privacy,

utility, and usability, ultimately enabling safer and more responsible use of AI in sensitive domains.

1.3 Aim and Objectives

Aim:

The intent of this paper is to develop a novel architectural framework for efficiently protecting personally identifiable information (PII) in Large Language Model inference while maintaining computational efficiency, semantic context, and model performance in privacy-sensitive applications.

Objectives:

Objective 1: Comprehensive Analysis of Existing Privacy-Preserving Methods

Systematically review current privacy-preserving techniques for LLM inference, including cryptographic approaches and differential privacy frameworks. Identify limitations, computational overhead, and privacy vulnerabilities, focusing on applicability to real-time chatbot applications in healthcare and financial domains.

Objective 2: Development of Client-Side PII Detection and Replacement Framework

Design and implement client-side Named Entity Recognition (NER) methodology for identifying and replacing PII data with contextually appropriate synthetic alternatives while preserving semantic meaning and conversational flow before server transmission.

Objective 3: Architecture Design and Performance Optimization

Propose a lightweight, scalable architectural pattern integrating PII protection mechanisms without compromising LLM inference efficiency. Optimize for client-side deployment, minimizing computational overhead while maintaining high accuracy in detection and performance.

1.4 Research Questions (IF ANY)

This study addresses three fundamental research questions that are critical to developing an effective privacy-preserving LLM inference framework:

RQ1: Client-Side NER Implementation Efficiency

How can Named Entity Recognition (NER) objects be effectively extracted on the client side while minimizing computational overhead and performance impact in real-time chatbot applications? This question examines the trade-offs between deploying lightweight NER frameworks versus client-side LLM-based extraction methods, with particular focus on processing speed, resource consumption, and accuracy in identifying personally identifiable information.

RQ2: Impact of NER-Based PII Replacement on Model Performance

To what extent does the replacement of identified PII entities with synthetic alternatives affect the accuracy and quality of LLM outputs? This investigation will quantify the performance degradation, if any, when original sensitive data is substituted with contextually appropriate fake data, and identify optimal replacement strategies that maintain semantic coherence.

RQ3: Privacy-Context Balance in Conversational AI

How can conversational context be preserved while ensuring complete privacy protection in multi-turn dialogues with LLMs? This question explores methods for maintaining semantic continuity and contextual understanding across conversation sessions without compromising the anonymization of sensitive information, particularly in domains requiring strict privacy compliance such as healthcare and financial services. These research questions collectively address the core challenge of developing a practical privacy-preserving framework that maintains both computational efficiency and conversational quality while ensuring robust protection of sensitive user data.

1.5 Scope of the Study

In this case study, the following points cover the scope of the study:

- Explain the Dataset used
- Explain Pre-processing done
- Study existing Models
- Finding out solution that doesn't compromise on Privacy and Speed
- Propose Architecture only covers PII data (like Contact details, Demographic information, Personal identifiers etc) protection and efficiency but doesn't cover other type of secret data like

- PHI (Protected Health Information) - Medical conditions, prescriptions, diagnoses.
- PIB (Personally Identifiable Behaviour)- Browsing history, app usage, search queries.
- Secrets or Credentials- API keys, passwords, access tokens, SSH keys.
- Financial Information - Bank balances, investment data, transaction details.
- Intellectual Property (IP) - Source code, internal documentation, trade secrets.
- National Security or Classified Data - Defence-related terms, classified project names.

1.6 Significance of the Study

Data breach costs hit a record peak of \$4.88 million in 2024, representing a 10% rise from the previous year. Approximately 46% of security incidents compromised customer personally identifiable information, encompassing tax identification numbers, email addresses, telephone numbers, and residential addresses. In 2024, ~275 million healthcare records were breached in the U.S (2024-healthcare-data-breach-report).

Financial institutions face an average data breach cost of roughly \$6.08 million per incident when personally identifiable information is compromised in 2025. Regulatory penalties for non-compliance may total up to 4% of worldwide annual revenue under GDPR, with other frameworks imposing thousands of dollars per individual violation. Approximately 50% of U.S. citizens have suffered personally identifiable information breaches within the past five years, as billions of data records face annual exposure. These financial calculations exclude difficult-to-measure consequences such as psychological trauma, credit rating deterioration, and missed opportunities stemming from PII misuse.

Proposed Architecture that could enhance the protection of PII data from client side chatbots whereby providing a more reliable and efficient way of preserving PII data. This would enhance the security and trust of LLM chatbots particularly amongst financial/ healthcare and education industry where protecting your PII data is utmost

important and is mandated by regulators. This contributes in potentially reducing frauds by not leaking PII data over internet.

1.7 Structure of the Study

This research is organized into six comprehensive chapters, each addressing a key aspect of the investigation into privacy-preserving mechanisms for Large Language Models (LLMs) using PII-aware processing frameworks.

Chapter 1 – Introduction

This chapter introduces the research background, motivation, and problem statement. It highlights the growing importance of privacy protection in AI-driven text generation, emphasizing risks associated with Personally Identifiable Information (PII) leakage. The chapter defines research objectives, hypotheses, and key contributions, and outlines the scope and limitations of the study.

Chapter 2 – Literature Review

The literature review surveys existing research on PII detection, anonymization, and data protection frameworks. It analyzes current trends in data privacy laws (e.g., GDPR, HIPAA), traditional masking techniques, and recent advancements in LLM-based PII handling. Comparative studies of existing anonymization and obfuscation methods are examined to identify knowledge gaps and motivate the proposed approach.

Chapter 3 – Research Methodology

This chapter details the research design, methodology, and experimental framework adopted for the study. It defines the overall approach, including dataset selection, preprocessing steps, evaluation metrics, and implementation tools. The conceptual design of the CleanSplit Model or equivalent privacy-preserving framework is elaborated, supported by system architecture diagrams and processing pipelines.

Chapter 4 –Implementation

This chapter provides a comprehensive overview of the dataset used (e.g., Kaggle PII dataset), including data characteristics, structure, and annotation procedures. It explains data preprocessing, labeling of PII entities, and integration of the dataset into the

experimental pipeline. Implementation details of each anonymization strategy—**Real Names (Baseline)**, **Fake Names Replacement**, **XXXX Masking**, and **LLM-based PII Removal**—are discussed with examples and configuration parameters.

Chapter 5 – Results and Evaluation

This chapter presents the empirical analysis and quantitative results. It covers:

Evaluation Metrics: Privacy, quality, and efficiency indicators such as PII Leakage Rate, Re-identification Risk, BLEU, ROUGE, Perplexity, and Coherence.

Experimental Procedure: Data preparation, execution pipeline, and statistical testing.

Results and Discussion: Comparative analysis of each method’s privacy protection effectiveness, linguistic quality, and efficiency trade-offs. Key findings are interpreted in the context of privacy–utility balance and their implications for secure LLM deployment.

Chapter 6 – Conclusion and Future Work??

The final chapter summarizes major findings, theoretical and practical contributions, and limitations of the study. It concludes with recommendations for applying privacy-preserving frameworks in real-world AI systems and proposes future directions, such as adaptive anonymization, federated PII learning, and integration of regulatory compliance mechanisms.

CHAPTER 2

LITERATURE REVIEW

Large Language Models (LLMs) and other advanced machine learning (ML) systems have become integral to countless applications, from personal assistants to healthcare diagnostics (Yu et al., 2022a) (Sun et al., 2025) (Asthana et al., 2025). Their power is derived from training on vast datasets, which frequently contain Personally Identifiable Information (PII) and other sensitive data. This dependency creates a significant privacy risk, as ML models have been shown to memorize and leak sensitive information from their training data, both during the training process and at inference time when interacting with users. These leakages can be exploited through various attacks, leading to breaches of individual privacy and exposing organizations to legal and reputational damage under regulations like GDPR and HIPAA (Lukas et al., 2023) (Johnson et al., 2020) (Asthana et al., 2025).

This review surveys the current landscape of PII data protection in machine learning. It begins by exploring the fundamental nature of privacy in the context of natural language and outlines the primary threats and attack vectors that lead to PII leakage. It then provides a comprehensive overview of the major categories of privacy-preserving techniques, including data-level sanitization, model-level and algorithmic defences like Differential Privacy and distributed learning, and cryptographic methods such as Homomorphic Encryption. Finally, it discusses the methodologies for evaluating these techniques and concludes by summarizing the key challenges and future research directions in this critical field.

2.1 Introduction

The proliferation of Large Language Models (LLMs) has led to their widespread adoption in conversational agents and chatbots across sensitive domains such as healthcare and customer service. While LLMs offer powerful capabilities, they inherently introduce critical privacy and security concerns. These concerns stem primarily from the models' tendency to **memorize and regurgitate** specific phrases, passages, or entire sensitive records from the data they process.

Privacy risks in chatbots extend beyond training data leakage to the **inference stage**, where user prompts themselves may contain Personally Identifiable Information (PII) or Protected Health Information (PHI). This sensitive input can be involuntarily retained by the model or actively exploited through various attack vectors, including training data extraction, PII extraction, reconstruction, and inference attacks. This body of work seeks to evaluate and compare strategies for mitigating these risks by transforming user input before it reaches the conversational AI, balancing the competing goals of *privacy protection* and *utility preservation*.

Research has demonstrated that LLM-powered chatbots, such as ChatGPT, are "highly prone to retaining and copying PII from the input when prompted". In one study, when prompted to summarize cover letters containing PII, the model retained PII verbatim in over half of the cases (57.4%). This involuntary leakage can occur even when the model is not malicious, simply due to its nature of memorizing data (Priyanshu et al., 2023)

Many studies have been conducted for coming up with a solution to anatomize the user inputs that contain PII sensitive data.

Differential Privacy

According to (Yang et al., 2022a) (Wei et al., 2020a) (Yang et al., 2022b) (Yu et al., 2022a) (Kerrigan et al., 2020), the implementation of a *generalized gradient perturbation split learning framework* introduces a robust mechanism for ensuring **differential privacy** within distributed and federated learning environments. Their studies collectively emphasize the necessity of preserving data confidentiality when multiple clients collaboratively train a shared model without exposing their local data. In this framework, gradient updates transmitted between local clients and the central server are perturbed with statistically calibrated noise. This technique provides formal **(ϵ , δ)-differential privacy** guarantees, ensuring that the presence or absence of any single user's data has a negligible influence on the overall model parameters or communication exchanges.

While this approach significantly enhances privacy protection, it introduces a well-documented **privacy–utility trade-off**. As observed by (Yang et al., 2022b) and (Kerrigan et al., 2020), achieving stronger privacy (represented by lower epsilon values)

necessitates injecting higher magnitudes of noise into the gradient signals. This process inevitably degrades model accuracy, slows convergence, and reduces the overall predictive reliability of the trained model. The impact of this degradation becomes even more pronounced when differential privacy is applied to **small-scale or sparse datasets**, where noise can obscure meaningful patterns to such an extent that the model's analytical capability is severely compromised (Wei et al., 2020a).

In the context of **chatbot and conversational AI systems**, these limitations are particularly restrictive. Chatbots often rely on relatively small, domain-specific, or personalized dialogue datasets, where linguistic nuances and contextual continuity are critical to system performance. The excessive noise introduced by differential privacy mechanisms can distort semantic representations, leading to incoherent responses, loss of contextual relevance, and reduced conversational fluency. Consequently, although differential privacy-based gradient perturbation frameworks are theoretically sound and highly suitable for large-scale systems such as financial or healthcare networks, their **practical applicability to dialogue-driven AI systems remains limited**.

To overcome these challenges, recent research has begun exploring **alternative privacy-preserving strategies** that maintain a better balance between confidentiality and model performance. These include **federated learning with selective gradient sharing**, **knowledge distillation-based privacy**, and **secure multi-party computation (SMPC)**.

Homomorphic Encryption Protection Analysis

Research by (Gilbert & Gilbert, 2024), (Bae et al., 2025a), (Liu & Liu, 2023) has highlighted **Homomorphic Encryption (HE)** as a promising cryptographic approach for ensuring robust data privacy in distributed and cloud-based computing environments. Unlike conventional encryption schemes that require data to be decrypted before computation, HE enables **direct mathematical operations on encrypted data** while preserving the confidentiality of the underlying information. This means that intermediate systems—such as cloud servers or third-party processors—can perform complex computations (e.g., addition, multiplication, or model inference) on ciphertexts without ever accessing the plaintext. As a result, sensitive datasets such as **personally identifiable information (PII)** or financial records remain fully encrypted throughout the entire data processing lifecycle, significantly reducing exposure risk and mitigating potential attack vectors.

Despite these strong theoretical advantages, homomorphic encryption faces substantial **practical limitations** that hinder its widespread adoption in real-world applications. The most significant challenge is its **computational inefficiency**. As noted by (Bae et al., 2025a), operations performed on encrypted data can be **up to 1,000 times slower** than equivalent operations on plaintext. This high computational overhead stems from the mathematical complexity of maintaining algebraic consistency across encrypted domains, often requiring intensive modular arithmetic and large ciphertext sizes. Consequently, HE-based systems demand **specialized hardware accelerators**, high memory capacity, and substantial energy consumption to achieve acceptable performance levels.

These constraints make **real-time processing of sensitive data**—such as continuous financial transactions, credit scoring, or medical monitoring—largely **impractical** without significant architectural optimization. Furthermore, implementing HE at scale introduces additional design complexities related to **key management**, **ciphertext expansion**, and **error accumulation**, which can further degrade system performance. Therefore, while homomorphic encryption provides **strong theoretical privacy guarantees**, its current computational burden limits its practicality for latency-sensitive applications like **banking analytics**, **fraud detection**, or **conversational AI systems** handling PII. Ongoing research aims to mitigate these issues through **partially homomorphic schemes**, **hardware-assisted acceleration** (e.g., GPUs, FPGAs), and **hybrid models** that combine HE with other privacy-preserving techniques such as **secure multi-party computation (SMPC)** or **differential privacy** to achieve a more balanced trade-off between security and efficiency.

Hide and Seek (HaS)

(Y. Chen et al., 2023) introduced an innovative framework aimed at safeguarding **prompt privacy** within **large language models (LLMs)**, addressing one of the most pressing concerns in the deployment of conversational AI systems—unintended exposure of sensitive or personally identifiable information (PII) embedded within user prompts. The proposed **HaS (Hide and Seek)** system integrates two complementary methodologies: **concealment of private entities through anonymization** and **retrieval of those entities via controlled de-anonymization**. In the anonymization phase, the framework employs specialized components to identify and mask sensitive entities (such as names, account numbers, or identifiers) before the prompt is processed

by the LLM. Subsequently, in the de-anonymization phase, the concealed entities can be selectively restored once the model’s output has been verified as privacy-safe, ensuring that private information remains protected throughout the inference process. The HaS architecture employs a **generative Bloomz model**—a multilingual open-source large language model—trained on data annotated by **GPT-4**, which provides high contextual sensitivity and entity-recognition accuracy. This dual-model setup enables dynamic anonymization and de-anonymization during real-time interaction, effectively reducing the risk of PII leakage through prompts or model responses. However, despite its conceptual robustness, the framework faces **significant practical limitations**. The incorporation of dual processing components and the reliance on large generative models substantially increase **computational overhead on the client side**, making chatbot deployments resource-intensive. Moreover, the **latency introduced** by the two-stage anonymization–de-anonymization pipeline hampers real-time conversational performance, particularly in low-power or latency-sensitive environments such as mobile and enterprise customer service applications. Consequently, while the HaS framework demonstrates strong potential for privacy preservation in LLM-driven systems, it currently struggles to achieve the **necessary balance between privacy assurance, computational efficiency, and user experience**, highlighting the need for lightweight yet effective prompt privacy mechanisms in production-grade chatbot architectures.

Split-N-Denoise Framework Analysis

(Mai et al., 2024) (Vepakomma et al., 2018) proposed a **privacy-preserving inference framework** that introduces a **client-side denoising model** to improve data utility following noise-based perturbation. The central premise of this approach lies in leveraging prior knowledge of raw input characteristics and the corresponding noise distribution to reconstruct more informative embeddings after the application of privacy-preserving noise. By performing denoising at the client level, the framework seeks to retain the functional accuracy of downstream tasks such as classification or prediction, even when strong **Differential Privacy (DP)** constraints are imposed during the data transmission or aggregation process. This hybrid strategy attempts to balance the competing objectives of maintaining privacy protection and ensuring model usability across distributed machine learning environments.

However, despite its innovative architecture, this framework exhibits **critical limitations in safeguarding Personally Identifiable Information (PII)**. The core of the approach still depends heavily on **Differential Privacy mechanisms**, which are fundamentally designed to protect *statistical aggregates or summary information* rather than *individual data instances*. As highlighted by (Vepakomma et al., 2018) and later reaffirmed by (Mai et al., 2024), the injected noise in DP-based systems can reduce the probability of identifying individuals but does not entirely eliminate the risk of **re-identification or membership inference attacks**, especially when adversaries possess auxiliary information.

This vulnerability becomes particularly concerning in **high-stakes domains such as healthcare and finance**, where regulations—such as HIPAA, GDPR, or the Digital Personal Data Protection (DPDP) Act—require **complete protection or anonymization of PII**. In such environments, even partial exposure of sensitive attributes (e.g., transaction IDs, patient identifiers, or demographic information) can have severe legal and ethical implications. Moreover, when applied to **sensitive textual data**, the stochastic noise introduced through DP can distort semantic content without fully obfuscating identifiable linguistic patterns, making de-anonymization theoretically possible. Consequently, while client-side denoising enhances the utility of differentially private embeddings, it **fails to achieve comprehensive PII protection**, underscoring the need for more robust hybrid privacy frameworks that combine DP with **encryption-based or secure multi-party computation (SMPC)** techniques to ensure both privacy and functional fidelity.

The Nature of Privacy and PII in Language Models

Protecting PII in language models is more complex than simply redacting direct identifiers like names or social security numbers (Lukas et al., 2023). The sources highlight a fundamental tension between technical data protection methods and the inherently social and contextual nature of privacy.

A core challenge is that "data protection is not equivalent to privacy protection". True privacy preservation requires respecting **contextual integrity**—understanding the norms and context in which information is shared—an ability current LLMs lack. Humans respect privacy not by failing to memorize information, but by judging the appropriateness of sharing it, a nuance that technical systems cannot yet replicate (Brown et al., 2022). Many privacy-preserving techniques are based on

assumptions that do not hold for natural language, where private information can be amorphous, embedded in prose, and shared among groups of individuals, blurring the boundaries of data ownership and sensitivity.

Furthermore, the very definition of a "leak" is complex. While many studies focus on **verbatim memorization**(Ippolito, n.d.), where a model exactly reproduces a string from its training set, this is an insufficient standard. Research shows that models can easily be prompted to produce **approximate memorization**—paraphrased or stylistically altered versions of sensitive text—which bypasses simple verbatim filters, creating a false sense of privacy. This suggests that a significant portion of privacy risks may be underestimated by current evaluation methods.

2.2 Privacy Threats and Attack Vectors

The leakage of PII can occur unintentionally through model behaviour or be actively exploited by malicious adversaries. The literature categorizes these threats into several key types(Jegorova et al., 2022).

- **Involuntary Leakage:** This occurs without malicious intent and is inherent to how models learn.
- **Memorization and Regurgitation:** LMs are prone to memorizing and reproducing sequences from their training data, particularly atypical or frequently repeated data points, which are more likely to be sensitive(Brown et al., 2022)(Jegorova et al., 2022). This risk extends to the inference stage, where models may copy, or **regurgitate**, PII from a user's input prompt directly into the output. Studies show that this regurgitation can be frequent and may even exhibit biases across different demographic subgroups(Priyanshu et al., 2023).
- **Malicious Attacks:** Adversaries can actively probe models to expose private data.
- **Data Extraction Attacks:** An attacker with API access can query a model to reconstruct verbatim or approximate training examples. These can be **untargeted**, aiming to recover any training data, or **targeted**, focusing on specific types of PII like email addresses or phone numbers.(Brown et al., 2022)(Huang et al., n.d.)^o (Lukas et al., 2023)
- **Reconstruction and Inference Attacks:** A more informed attacker can use a sentence with masked PII (e.g., "John Doe lives in [MASK]") to have the model

reconstruct the missing information. This is even more effective if the attacker has a list of potential candidates (PII inference). Advanced attacks can leverage both the prefix and suffix of the masked context to improve success rates by over 10x compared to simpler methods.(Lukas et al., 2023)

- **Indirect Prompt Injection:** This attack exploits a fundamental flaw in LLMs: their inability to reliably separate instructions from data. An attacker can embed malicious commands within a piece of text meant for processing (e.g., an email to be summarized). The LLM may then execute this hidden command, potentially leading it to leak PII or perform other unauthorized actions.(Zverev et al., 2025)

2.3 Privacy-Preserving Techniques for PII Protection

A diverse array of techniques has been developed to mitigate PII leakage. These can be broadly categorized into data-level, model-level, and cryptographic approaches.(Jegorova et al., 2022)

- **Data-Level Techniques: Sanitization and Anonymization** These methods involve removing or replacing sensitive information before it is processed by a model.
- **NER-Based De-identification:** A common approach is to use Named Entity Recognition (NER) models, such as fine-tuned BERT, to identify and scrub PII from text. However, this method is imperfect; NER models have limited recall, meaning some PII is inevitably missed, and performance degrades significantly when applied to new data domains due to differing annotation standards and data distributions.(Lukas et al., 2023)(Johnson et al., 2020)
- **Client-Side Prompt Sanitization:** For API-based LLMs where the model itself cannot be modified, sanitization can be performed on the client's device before the prompt is sent to the server. Frameworks like **PP-TS** (Kan et al., n.d.), **Casper** (Chong et al., 2024) , and **Rescriber** (Zhou et al., 2025) implement this. They range from fully automated multi-layered systems combining rules, NER, and local LLMs for topic detection (Casper), to user-in-the-loop tools that empower users to make final decisions on redaction or abstraction (Rescriber). These tools enhance user control and can have an educational effect by raising privacy awareness. A key challenge is ensuring sanitization does not

compromise the utility of the LLM's response, which some frameworks address with "reasonability checks" (Kan et al., n.d.).

2.3.1 Algorithmic Techniques: Distributed Learning and Differential Privacy

These techniques modify the model training or inference process to protect privacy.

- **Distributed Learning:** The core idea is to keep raw data decentralized on user devices.
- **Federated Learning (FL)** enables multiple clients to collaboratively train a model without sharing their local data; only model updates (e.g., gradients or weights) are sent to a central server for aggregation (Wei et al., 2020b). Combining FL with **Parameter-Efficient Fine-Tuning (PEFT)**, such as LoRA, makes it feasible to tune large models on resource-constrained devices, a paradigm known as **Federated Instruction Tuning (FedIT)** (Zhang et al., 2024). In this setting, the heterogeneity of data across clients becomes an advantage, improving the robustness and generalizability of the global model.
- **Split Learning (SplitNN)** splits the neural network architecture between the client and the server. The client processes data through the initial layers and sends only the intermediate activations to the server, which completes the computation (Vepakomma et al., 2018). This approach can be more computationally and communication-efficient for the client than FL in certain scenarios, such as when model sizes or the number of clients is large (Singh et al., 2019). Privacy can be further enhanced by adding noise to the intermediate activations, with frameworks like **Split-N-Denoise (SnD)** using a client-side denoising model to recover utility from the server's noisy output .
- **Differential Privacy (DP):** DP is the "gold standard" for providing mathematically rigorous privacy guarantees by adding calibrated noise to data or computations (Li et al., 2022)(Kumar et al., n.d.).
- **DP-SGD** is the primary algorithm for training neural networks with DP, which involves clipping the norm of per-example gradients and adding noise before aggregation (Yu et al., 2022a) (Jegorova et al., 2022).

Applying DP to large models is challenging due to high computational costs and utility degradation. However, combining DP-SGD with **PEFT methods like LoRA** has proven highly

effective for privately fine-tuning LLMs. This approach significantly reduces overhead and can lead to private models whose performance approaches that of their non-private counterparts. Surprisingly, studies show that larger models can be more robust to DP noise and experience a smaller relative drop in accuracy when fine-tuned this way.

Despite its strong guarantees, DP is not a panacea. Its record-level privacy definition is a poor fit for interconnected textual data, and it has been shown to still permit a non-trivial amount of PII leakage, especially for information that appears frequently across the dataset.

2.3.2 Cryptographic Techniques

Homomorphic Encryption (HE) HE enables computation directly on encrypted data, offering a powerful way to outsource processing to untrusted servers without revealing plaintext information (Gilbert & Gilbert, 2024a).

- **HE for Private Inference:** A major challenge is that HE schemes typically only support polynomial operations (addition and multiplication), while transformers contain complex non-polynomial functions like GELU, softmax, and Layer-Norm (T. Chen et al., 2022). The **THE-X** framework addresses this by approximating these functions with HE-friendly alternatives (e.g., replacing GELU with ReLU) and using a hybrid client-server protocol where the client performs the non-polynomial steps on temporarily decrypted data.
- **HE for Private Retrieval:** For retrieval-augmented LMs, HE can be used to build an encrypted vector database. This allows a client to perform secure semantic search over a private, remote database, retrieving relevant documents without the server learning anything about the query or the data content (Bae et al., 2025a).
- The primary limitation of HE is its significant performance overhead in computation and communication, which currently restricts its use in many real-time applications (Gilbert & Gilbert, 2024a).

2.4 Evaluation Methodologies and the Privacy-Utility Trade-off

Evaluating the effectiveness of PII protection techniques requires balancing two competing goals: privacy and utility. A central finding across the literature is the existence of a privacy-utility trade-off: stronger privacy protections, such as adding more noise in DP, generally lead to a degradation in model performance (Yu et al., 2022a).

To systematically measure this trade-off, dedicated evaluation frameworks have been proposed. The "Privacy-preserving4LLM Benchmarking" framework, for instance, provides a methodology for assessing algorithms across different real-world scenarios (e.g., protecting training data vs. user queries) and uses a Parameter Optimizer to ensure fair comparisons (Sun et al., 2025).

- **Privacy Metrics:** Privacy is quantified using several methods. The exposure metric involves inserting "canary" data—unique, sensitive sequences—into a dataset and measuring how easily they can be extracted from the trained model (Sun et al., 2025). Other methods rely on the success rate of the attacks described in Section 3, such as the number of PII's successfully extracted or the accuracy of reconstruction attacks (Huang et al., n.d.).
- **Utility Metrics:** Utility is typically measured using standard NLP evaluation metrics on downstream tasks, such as perplexity for language modelling, accuracy and F1-score for classification, and BLEU or ROUGE scores for text generation (Kerrigan et al., 2020).

Intuitive approaches, such as anonymizing sensitive terms before LLM input, prove insufficient for comprehensive privacy protection. These methods fail to conceal other linguistic elements vital for semantic interpretation, thereby compromising both privacy and the effectiveness of tasks requiring precise semantic understanding. Additionally, existing denoising techniques face fundamental limitations when deployed server-side, as the server lacks knowledge of injected noise levels, creating an inherent conflict with privacy protection objectives.

2.5 Research Gap and Contribution

The literature reveals a persistent and critical gap in the development of privacy-preserving methodologies capable of simultaneously ensuring robust protection of individual PII and maintaining the semantic integrity and performance of large language models (LLMs). Existing approaches—such as those based on Differential Privacy, Homomorphic Encryption, or client-side denoising frameworks—tend to exhibit a trade-off between privacy strength and data utility. Methods offering stronger privacy guarantees often degrade model accuracy and fluency, while utility-preserving techniques fail to provide sufficient protection for fine-grained, individual-level

information. Consequently, current privacy-preserving solutions remain inadequate for high-sensitivity applications, particularly in sectors like healthcare and financial services, where conversational AI systems process context-rich, personally identifiable data under stringent regulatory constraints (e.g., GDPR, HIPAA).

To address these limitations, this study proposes an enhanced architecture termed the CleanSplit Model (CSM). The CSM framework introduces a hybrid privacy-preserving mechanism that integrates Named Entity Recognition (NER) with context-aware data synthesis. In this architecture, NER algorithms are employed to detect and classify PII entities—such as names, contact details, or financial identifiers—within user inputs. Identified entities are then replaced with synthetically generated, semantically aligned placeholders, ensuring that the contextual flow and intent of the conversation remain intact. Unlike traditional anonymization or noise-based approaches, CSM preserves both linguistic coherence and domain relevance, thereby minimizing the trade-off between privacy protection and model performance.

By maintaining contextual fidelity while guaranteeing complete PII masking, the CleanSplit Model represents a promising direction for developing trustworthy, privacy-compliant chatbot systems. Its design directly supports deployment in regulatory-intensive domains, such as healthcare diagnostics, financial advisory services, and digital customer support—where safeguarding user trust and adhering to compliance standards are as crucial as model accuracy and responsiveness.

2.6 Conclusion and Future Directions (DO we need it here?)

The protection of PII in machine learning is a complex and rapidly evolving field. This review has highlighted the multifaceted nature of privacy threats, from unintentional model memorization to sophisticated adversarial attacks, and surveyed the three main pillars of defense: data-level sanitization, algorithmic approaches like DP and distributed learning, and cryptographic methods like HE.

While significant progress has been made, no single technique offers a complete solution. Data sanitization is practical but often superficial and brittle. Distributed learning architectures like FL and SplitNN are powerful for keeping data local but still expose model updates or activations that can leak information. Differential Privacy

offers provable guarantees but comes at a cost to utility and is based on assumptions that are a poor fit for natural language. Homomorphic Encryption provides strong security but remains computationally expensive for widespread use.

The most promising solutions appear to be hybrid approaches. Parameter-efficient fine-tuning has made DP practical for large models. Client-side tools that combine multiple sanitization techniques and empower users with control are gaining traction. Cryptography is finding a role in securing specific components of larger systems, like retrieval databases.

Key challenges remain. The community must move beyond simplistic, verbatim definitions of memorization to address approximate or semantic leakage (Ippolito, n.d.). The fundamental lack of instruction-data separation in LLMs poses a systemic risk that may require architectural innovations to solve (Zverev et al., 2025). Finally, bridging the gap between technical data protection and the rich, contextual nature of human privacy remains the ultimate goal and a central open question for future research (Brown et al., 2022).

Table 2.1: of Various papers used in Literature review starting from the latest first

Paper	Methodology Used	Outcome	Shortcomings / Future Research
(Bae et al., 2025b).	The methodology employs a hybrid framework combining Socratic Chain-of-Thought Reasoning with a Homomorphically Encrypted Vector Database	The hybrid setup of remote and local LLMs enhances privacy and boosts performance	Future work should target tasks resistant to clean decomposition,
(Gilbert & Gilbert, 2024b)	The study reviews HE schemes, analyzes their computational efficiency, and simulates secure models for privacy-preserving deep learning and multiparty computation.	Homomorphic Encryption is crucial for secure computation, but future work must address its performance, scalability, and integration challenges.	Homomorphic Encryption faces high overhead, slow performance, resource demands, key management issues, limited operations, and poor platform support.
(Sun et al., 2025)	The framework includes a Protection Scenario Manager, a Parameter Optimizer for utility-privacy balance, and a Privacy-Level Evaluator using canary exposure, analyzing three DP algorithms across multiple LLMs and datasets.	CusText+ offers optimal utility-privacy balance, Mahalanobis ensures strongest privacy, Llama2-7b is compatible, Mistral-7B resilient, and Falcon-7b vulnerable.	Current privacy-preserving algorithms trade utility for protection, prompting future work on tailored attacks, algorithms, and latency effects
(Zhou et al., 2025)	Rescriber is a browser extension using on-device LLMs for user-led data minimization via detection, sanitization, and privacy recovery, evaluated with 12 users comparing Llama3-8B and GPT-4o against Presidio.	Rescriber empowered user-driven data minimization and showed Llama3-8B performs comparably to GPT-4o in usability tests.	The study highlights privacy misperceptions, abstraction gaps, and utility loss, urging better small LLMs and placeholder strategies.
(Mai et al., 2024)	The Split-N-Denoise (SnD) method safeguards privacy by splitting LLM inference—adding Laplacian noise to local embeddings before cloud processing and denoising outputs locally using a transformer-based model for improved utility.	The Split-N-Denoise (SnD) framework achieves over 10% higher utility than DP-based baselines while preserving strong local privacy, maintaining performance even under extremely low privacy budgets ($\eta \leq 0.01$) across diverse LLMs and tasks.	The Split-N-Denoise (SnD) framework faces scalability and computational challenges with large LLMs, requiring lighter denoising methods and improved accuracy; future work aims to extend it to Sequence-to-Sequence inference while mitigating noise amplification issues.
(Chong et al., 2024)	The framework combines rule-based filters, ML-based NER, and a local LLM to sanitize sensitive data with high accuracy and low overhead.	The Casper extension enables client-side prompt sanitization with 98.5% PII filtering and 89.9% sensitive topic detection, ensuring efficient privacy for LLMs.	The paper identifies English-only support, workflow friction, and detection ambiguity, with future work targeting multilingual, WebGPU, and trusted computing.
(Zhang et al., 2024)	FedIT enables privacy-preserving LLM fine-tuning via LoRA-based PEFT, sharing lightweight updates across 100 clients using partitioned Dolly data.	FedIT (Shepherd-7B) surpasses baselines by leveraging diverse client data for better generalization, even	FedIT still lags behind centralized models, with future work focusing on advanced federated optimization to better harness

		outperforming ChatGPT on some tasks..	client data diversity and close the performance gap.
(Zverev et al., 2025)	The study defines and measures instruction-data separation in LLMs using an empirical separation score with "surprise witnesses," evaluated on the SEP dataset to assess mitigation techniques like prompt engineering and fine-tuning.	Experiments show LLMs lack clear instruction-data separation; while prompt engineering and fine-tuning improve separation, they reduce utility, suggesting the need for architectural changes for robust mitigation.	The paper notes limitations in using proxy prompts and highlights persistent failures even in advanced LLMs; future work calls for new architectures and training methods to achieve true instruction-data separation and improved explainability.
(Kumar et al., n.d.)	This paper uses a systematic review to study AI's impact on data protection, covering literature search, selection, thematic synthesis, and quantitative analysis of regulatory and ethical trends.	The paper synthesizes literature and empirical findings to clarify AI's impact on data protection, highlighting GDPR, ethical principles, and technologies like federated learning and differential privacy to guide policymakers and researchers	The paper notes limitations like selection bias, English-only sources, and evolving AI/data protection; future work should pursue longitudinal studies, interdisciplinary ethics, privacy tech, algorithmic bias, and adaptive regulations.
(Huang et al., n.d.)	The paper evaluates privacy risks in kNN-LLMs using PII and training data attacks, measuring leakage via PII counts and ROUGE-L, and mitigating with sanitization, encoder decoupling, data mixing, and DP-SGD, while tracking utility through perplexity.	kNN-LLMs leak more private data than parametric models; sanitization and encoder decoupling mitigate targeted risks, public datastore and mixed fine-tuning help untargeted attacks, and higher k boosts both utility and privacy.	The findings mainly apply to kNN-LLMs, with future work needed on models like RETRO, broader datasets, and diverse privacy techniques to tackle untargeted risks.
(Y. Chen et al., 2023)	The Hide and Seek (HaS) framework trains local Hide and Seek models for anonymization and de-anonymization, using NER for label-based hiding, anonymizes user prompts locally, sends them to cloud LLMs, and restores sensitive data post-response.	HaS ensures prompt privacy by anonymizing inputs and restoring outputs using lightweight local models, showing strong utility and resistance to decryption in translation and classification tasks	HaS offers limited privacy by anonymizing entity words only, making it unsuitable for tasks needing precise semantics.
(T. Chen et al., 2022)	THE-X enables transformer inference on encrypted data via homomorphic encryption by approximating non-polynomial functions into HE-compatible operations with minimal performance loss.	THE-X enables encrypted transformer inference with theory-guaranteed privacy, minimal performance drop (1.48% GLUE, 1.9% CONLL2003), and full HE-compatible function conversion.	THE-X tackles the challenge of ciphertext inference in transformers by replacing unsupported computations with HE-friendly approximations, requiring LLM retraining; future work aims to improve approximations across NLP tasks.
(Kan et al., n.d.)	PP-TS uses three modules: local sanitization of sensitive data with semantic checks and mapping, remote LLM inference on sanitized input, and local recovery of original privacy info for seamless interaction.	PP-TS achieved strong privacy protection with 95.96% PRR, 92.33% DUR, and high DPR against attacks, maintaining model performance.	PP-TS may slightly degrade event extraction performance, but the reasonability check helps mitigate this; no major future work is noted.
(Ippolito, n.d.)	MEMFREE decoding filters n-grams via Bloom filters to block verbatim memorization, yet style-transfer prompts still leak data through approximate matches.	MEMFREE decoding blocks exact matches, but semantically similar outputs from models like GPT-3 and Copilot still leak data—highlighting the need for broader memorization definitions.	The paper critiques BLEU-based memorization thresholds and calls for semantic metrics, plagiarism tools, and clearer generalization boundaries.

(Liu & Liu, 2023)	The methodology enhances privacy-preserving transformer inference via HE and MPC by replacing costly operators with efficient alternatives and fine-tuning layer-wise to retain accuracy.	The paper achieves 5x faster computation and 80% less communication in private transformer inference by replacing costly operators with efficient approximations, maintaining or improving accuracy.	Private inference remains costly vs plaintext, limited by model size and semi-honest security; future work targets pruning, distillation, hardware acceleration, malicious security (e.g., ZKP), and early LayerNorm analysis.
(Priyanshu et al., 2023)	The methodology evaluates ChatGPT’s privacy risks by testing input regurgitation and prompt-induced sanitization using synthetic PHI/PII datasets and leakage metrics.	ChatGPT regurgitated PII in 57.4% of cases, but prompt-induced sanitization reduced leakage to 30.5% with general and 15.2% with step-by-step guidance.	Prompt-induced sanitization has limitations, showing subgroup bias in PII leakage and reducing utility of non-sensitive data.
(Lukas et al., 2023)	The NbAFL methodology perturbs client updates with noise before aggregation, analyzes convergence-privacy trade-offs, proves DP guarantees, explores K-client scheduling, and validates results via MLP simulations.	NbAFL satisfies DP with Gaussian noise, reveals convergence-privacy trade-offs, identifies optimal T and K, and is validated via simulations	A key shortcoming is the lack of analysis on NbAFL’s convergence with varied client data; future work aims to address this.
(Yu et al., 2022b)	The framework fine-tunes frozen LLMs using DPSGD on minimal new parameters via LoRA-like methods, ensuring privacy and efficiency	The framework achieves top privacy-utility tradeoffs, showing larger LLMs like RoBERTa-Large and GPT-2-XL retain accuracy under DP.	A key shortcoming is the unexplained resilience of DPSGD noise in full fine-tuning; future work aims to clarify this and explore private-efficient tuning
(Brown et al., 2022)	The paper critiques assumptions in sanitization and DP, arguing only public-purpose text ensures true LLM privacy.	The paper argues that sanitization and DP fail to ensure privacy for LMs and recommends training only on explicitly public text.	The paper finds sanitization and DP inadequate for LMs due to contextual, multi-user data, and proposes private personalization with public-only pretraining and local fine-tuning.
(Yang et al., 2022a)	The adaptive framework mitigates PII/SPI risks in LLMs via policy translation, contextual detection, and tailored anonymization, validated against benchmarks.	The adaptive framework achieved 0.95 F1 for Passport Numbers, 4.6/5 user trust, and validated GDPR-compliant anonymization.	Shortcomings include evolving laws, jurisdictional conflicts, and cross-lingual PII detection; future work targets regulation reconciliation, domain expansion, and automation.
(Jegorova et al., 2022)	The paper surveys inference-time data leakage in ML models, categorizing leakage types, reviewing defenses, identifying gaps, and analyzing task-specific risks	The paper surveys inference-time leakage in ML models, categorizes leakage types, reviews defenses, and outlines future research challenges.	Shortcomings include uneven attack coverage, inefficient defenses, and unverifiable forgetting; future work targets scalable defenses and standardized frameworks.
(Yang et al., 2022a)	The paper introduces Marvell to mitigate label leakage in split learning by optimizing noise against gradient-based attacks under utility constraints.	The paper shows label leakage via gradients in split learning and introduces Marvell to optimize noise for privacy-utility balance.	The paper reveals label leakage via gradients in split learning, urging future work on complex threats, multiclass tasks, and multi-party setups.
(Kerrigan et al., 2020)	The paper fine-tunes a public base model on private data using DPSGD with noise and clipping, evaluated via perplexity.	DP fine-tuning of a public base model enables effective private language modeling, outperforming direct DP training on private data.	DP training on private data alone yields poor performance and incoherent outputs due to weak architectures and high perplexity.

(Johnson et al., 2020)	The paper uses BERT-based NER for medical text deidentification, fine-tuned on harmonized datasets and evaluated for sensitivity.	The BERT-based model achieved state-of-the-art PHI removal, balancing sensitivity and false positives, with open-source code.	Shortcomings include annotation ambiguity, poor portability, and BERT's numeracy limits; future work urges hybrid models and PHI standardization.
(Wei et al., 2020b)	NbAFL adds DP noise before FL aggregation, proves privacy guarantees, analyzes convergence tradeoffs, and optimizes client scheduling	NbAFL ensures DP in FL by adding noise pre-aggregation, optimizing convergence via client count, aggregation rounds, and K-client scheduling.	The paper fills a gap in DP-FL theory by analyzing privacy-convergence tradeoffs and suggests future evaluation across data distributions.
(Singh et al., 2019)	The paper compares communication efficiency of split vs federated learning by analyzing data transfer per client per epoch.	The paper shows split learning is more efficient with more clients or larger models, while federated learning excels with more data and fewer clients.	Further analysis of split vs federated learning should include resource usage and convergence epochs across practical scenarios.
(Vepakomma et al., 2018)	The paper proposes SplitNN configurations for healthcare that enable collaborative training without sharing raw data or model details.	SplitNN enables privacy-preserving healthcare model training with higher accuracy, lower client computation, and better communication efficiency.	The paper calls for advanced SplitNN designs and compression techniques to support diverse, efficient distributed learning across domains.

CHAPTER 3

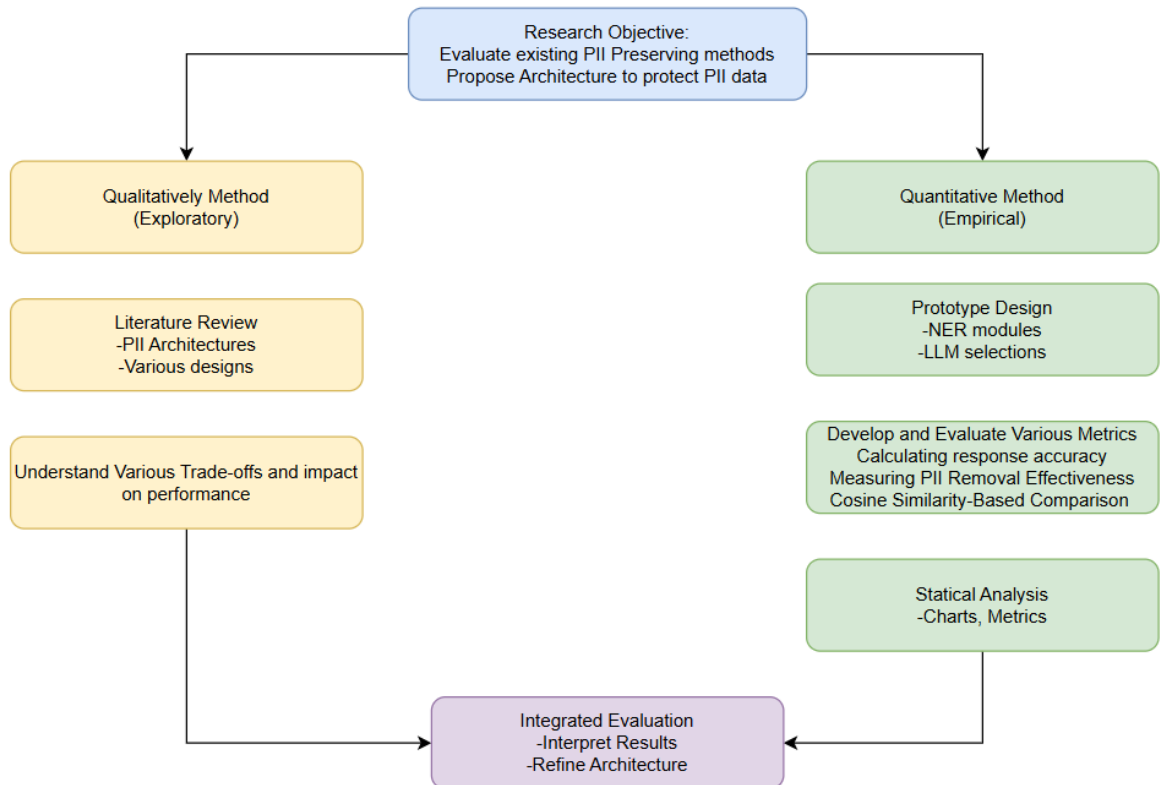
PROPOSED METHODOLOGY

This thesis investigates the effectiveness of different data sanitization strategies applied to user prompts in conversational AI systems (chatbots) to mitigate the risk of Personally Identifiable Information (PII) leakage at the inference stage. The methodology outlines the experimental design, the four input preparation conditions, the remote and local language models used, and the quantitative metrics employed to assess the resultant privacy-utility trade-off.

3.1 Approach

This study employs a mixed-methods research approach that combines both quantitative and qualitative methodologies to comprehensively investigate.

Figure 3.1: Displaying Research Methodology



3.1.2 Qualitative Component:

The qualitative component facilitates an in-depth exploration of the architectural design decisions underpinning the proposed CleanSplit Model (CSM) and their consequent influence on chatbot behavior and privacy performance. This phase emphasizes understanding the design rationales, implementation trade-offs, and contextual factors that shape system development choices. Through systematic evaluation of these aspects, the qualitative analysis aims to uncover how specific architectural elements—such as the integration of Named Entity Recognition (NER), context-preserving data synthesis, and privacy modules—affect the model’s interpretability, usability, and compliance readiness.

Furthermore, this component investigates developer and user perspectives on the architecture’s transparency, perceived trustworthiness, and adaptability within real-world applications such as healthcare and finance. Semi-structured observations and design documentation analyses are employed to derive thematic insights into how privacy-preserving features alter conversational dynamics, including user engagement, response fluency, and error tolerance. The resulting insights provide a conceptual understanding of how privacy-centric architectural decisions translate into observable chatbot performance and user experience outcomes, thereby informing future refinements to both model design and deployment strategies.

3.1.3 Quantitative Component:

The quantitative component is designed to facilitate the systematic and objective measurement of the CleanSplit Model’s (CSM) performance across multiple dimensions, including response accuracy, latency, computational efficiency, and the effectiveness of PII removal. This component emphasizes empirical validation through controlled experiments and statistical evaluation, ensuring that both functional and privacy-oriented performance aspects are rigorously assessed.

Key performance indicators (KPIs) are derived from standardized evaluation protocols commonly applied in privacy-preserving natural language processing (NLP) research. These include metrics such as precision, recall, and F1-score for response generation accuracy; average response time for latency analysis; and computational throughput for assessing resource efficiency. To evaluate privacy protection capability, specialized PII

leakage metrics and privacy loss scores (ϵ -values) are employed to quantify the extent of sensitive data exposure under various adversarial scenarios.

3.2 Experimental Design and Threat Model

The experimental design aims to empirically evaluate the CleanSplit Model (CSM) under realistic operational conditions that simulate privacy-sensitive chatbot interactions. This section outlines the threat scenario, experimental variables, and methodological considerations used to assess the model’s effectiveness in preventing PII leakage while maintaining high conversational performance.

3.2.1 Threat Scenario

The core threat addressed in this study is personally identifiable information (PII) leakage during the inference stage of large language model (LLM)-based conversational systems. In typical user interactions, individuals may willingly or inadvertently disclose sensitive information—such as names, identification numbers, account details, or medical data—within their queries to a chatbot. Once processed, this data can be memorized, cached, or semantically retained within the model’s temporary memory or attention mechanisms, leading to potential unintentional disclosure in subsequent responses.

Unlike traditional data privacy risks that arise during the training phase, this threat specifically manifests during inference, where user inputs are dynamically processed in real time. This distinction highlights a critical and underexplored privacy vulnerability in deployed conversational systems. Accordingly, the focus of this experiment is on input protection, also referred to as Query Data Protection (QDP) — a pragmatic defence strategy in which sensitive input data is identified, transformed, or anonymized before being transmitted to the model. This approach directly reflects real-world application scenarios in which user privacy must be safeguarded during active system interactions, such as in healthcare consultations, banking assistance, or customer service chatbots.

3.2.2 Experimental Variables

The experimental framework is structured around two primary variables that govern the evaluation design:

1. **Independent Variable** – Input Transformation Strategy:

This variable represents the privacy-preserving mechanism applied to the user input prior to model processing. The study investigates four distinct experimental conditions, representing a continuum from minimal to advanced protection:

Condition A: Raw input without transformation (baseline).

Condition B: Simple anonymization through keyword masking.

Condition C: NER and Faking (Converting PII data to Fake data)

Condition D: CleanSplit Model (CSM) employing Local LLM Conversion (Converting PII data using local LLM like Ollama)

2. **Dependent Variable** – Output Quality and Utility:

This variable captures the overall performance and usability of the chatbot's responses following each input transformation strategy. Key assessment criteria include semantic coherence, context preservation, user satisfaction, and response fluency, as evaluated through both automated and human-centered metrics.

By manipulating the input transformation strategy and measuring its downstream effects on output quality, the experimental design seeks to quantify the privacy–utility trade-off inherent in privacy-preserving conversational AI. The resulting comparisons provide empirical insight into the extent to which the proposed CleanSplit Model achieves a balance between PII protection, system accuracy, and user experience.

3.2.3 **Experimental Setup**

The experimental setup aims to evaluate the proposed CleanSplit Model (CSM) across different levels of privacy protection and contextual fidelity within chatbot interactions. The experimental design emulates a realistic environment where user queries containing PII are processed through a combination of local preprocessing modules and remote LLM-based conversational systems. The following subsections describe the configuration of the target chatbot, PII detection tools, and input transformation conditions used for evaluation.

3.2.3.1 Target Chatbot (Remote LLM)

The core conversational agent—representing an untrusted commercial service provider—is a state-of-the-art closed-source LLM accessed via an API (e.g., ChatGPT or an equivalent proprietary model). This remote model is assumed to operate within a cloud-hosted environment, with limited or no transparency regarding its internal operations or data-handling mechanisms.

The underlying threat model presumes that such commercial LLM providers may not share the same incentives for privacy protection as end users or organizations deploying them. Consequently, the model is treated as a black-box entity, where user prompts and responses could be logged, cached, or used for retraining. This setting reflects realistic privacy concerns in enterprise and consumer chatbot deployments, where users often have limited assurance of how their data is managed once transmitted to a cloud-based service.

3.2.3.2 PII Detection Tools

Experiments 2 through 4 employ explicit PII recognition mechanisms using Named Entity Recognition (NER)-based tools. PII detection and tagging are conducted using modules such as Microsoft Presidio and spaCy, which are widely adopted for text anonymization and data sanitization in privacy-sensitive NLP workflows. In these experiments, NER serves as the foundation for identifying sensitive entities—such as personal names, financial identifiers, addresses, or contact numbers—that require redaction or transformation.

In Experiment 4, a local generative model (simulated using Ollama) is introduced to enhance privacy protection. This lightweight, decoder-only language model is deployed locally on the user’s terminal, functioning independently of cloud-based LLMs. The local LLM facilitates on-device preprocessing of user queries, performing context-aware abstraction and rewriting before communication with the remote model. This configuration inherently strengthens privacy by keeping raw, sensitive data within the user’s computing environment, thereby minimizing data exposure to third-party cloud providers and enhancing user trust and control.

3.2.3.3 Input Transformation and Experimental Conditions

The experimental framework evaluates **four input transformation strategies**, each representing a progressively stronger privacy defense mechanism against PII disclosure during chatbot interaction.

Condition 1: Baseline (Sending PII data as is)

Mechanism: Raw user prompts containing PII are transmitted directly to the remote LLM without any preprocessing or sanitization.

Purpose: Serves as the **undefended baseline**, quantifying the inherent risk of **input regurgitation**—a phenomenon where chatbots reproduce sensitive data verbatim. Prior research has shown that LLMs can inadvertently replicate or recall user-provided identifiers in over half of evaluated cases, underscoring the severity of this threat.

Condition 2: NER and Masking (Sending PII data as masked)

This condition implements explicit scrubbing using standardized NLP techniques.

Mechanism: PII is identified via NER and replaced with a generic placeholder token, such as [MASK], or a general entity tag.

Rationale: PII scrubbing is a widely used dataset curation technique to remove PII. Masking maximizes privacy but risks severe loss of utility by removing critical contextual information necessary for the LLM to generate a useful response.

Condition 3: NER and Faking (Converting PII data to Fake data)

This condition utilizes substitution, a form of data sanitization or obfuscation.

Mechanism: PII is identified via NER and replaced with **dummy text** or a fixed, realistic-looking synthetic detail of the same type (e.g., a phone number replaced with "123-456-789"). This is also known as using **pseudo names**.

Rationale: Faking data aims to mitigate the risk of PII extraction by replacing real sensitive information with non-sensitive substitutes. This strategy is sometimes adopted by users intentionally to mask their identity.

Condition 4: Local LLM Conversion (Converting PII data using local LLM like Ollama)

This condition employs advanced, contextual abstraction through a local model.

Mechanism: The input is processed by a local LLM (e.g., a small Llama model via Ollama) to convert specific PII into generalized or **abstracted** language. This process

is a privacy-preserving text sanitization technique based on rewriting the input text. This approach aims to provide contextual information while avoiding specific, identifiable details.

Rationale: This method seeks to achieve a better balance between privacy and utility by performing linguistic generalization. It attempts to maintain semantic consistency, avoiding issues caused by simple masking or faking that might lead the remote LLM to misinterpret the query. The local model can act as a component to verify and rectify the plausibility of the rephrased text.

Collectively, these four experimental conditions enable a comprehensive comparison of **privacy-utility trade-offs** across varying levels of protection, ranging from unaltered raw data transmission to contextually rich, locally abstracted input transformation. The results from these conditions inform the empirical assessment of the **CleanSplit Model’s efficacy** in achieving optimal **PII protection, response coherence, and computational efficiency**.

3.2.3.4 Evaluation Metrics

The experimental evaluation focuses on quantifying the **privacy–utility trade-off** inherent in privacy-preserving conversational systems. Privacy is assessed implicitly through **input transformation effectiveness** (i.e., the extent of PII removal or protection), while utility is defined as the **preservation of information** necessary for the chatbot to fulfill the user’s original intent. Together, these measures enable a balanced assessment of how well the **CleanSplit Model (CSM)** safeguards sensitive data without compromising the model’s performance or usability.

3.2.3.5 Task-Dependent Performance Metrics

Because response evaluation depends on the type of downstream task, different quantitative metrics are employed for varying chatbot objectives:

(a) Classification or Structured Outputs

For tasks with a fixed answer set (e.g., yes/no, sentiment, or intent classification):

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}}$$

(b) Extractive or Factual Question Answering

For knowledge-based or fact retrieval tasks, two complementary metrics are used:

- **Exact Match (EM):** Proportion of outputs that exactly match the ground truth.
- **F1 Score:** Harmonic mean of precision and recall over overlapping tokens between the predicted and reference answers.

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

These metrics collectively reflect factual correctness and linguistic precision in LLM-generated responses.

(c) Safety and PII-Related Tasks

For privacy evaluation tasks, correctness is defined as the **absence of PII in the model output**.

$$\text{Response Accuracy} = \frac{\text{Responses with No PII Leakage}}{\text{Total Responses}}$$

PII detection tools such as **Microsoft Presidio**, **spaCy**, or **LlamaGuard** are used to automatically flag potential PII in generated outputs.

3.2.3.6 Measuring PII Removal Effectiveness

Evaluating PII protection involves systematically measuring how effectively sensitive data is removed or transformed before model inference. The procedure includes four key stages:

Step 1 — Define PII Categories:

PII entities are standardized using established taxonomies from **NIST**, **GDPR**, or **HIPAA** guidelines. Common entity types include personal names, phone numbers, addresses, SSNs, IPs, and usernames.

Step 2 — Select or Build Detection Tools:

PII recognition is performed using off-the-shelf or custom-built tools such as:

Microsoft Presidio

spaCy + regex or domain-specific NER models

AWS Comprehend / Google DLP

Custom LLM-based classifiers

These detectors tag PII entities in both **original** and **sanitized** text samples.

Step 3 — Compute Privacy Metrics:

(A) Precision, Recall, and F1 Score (when ground truth exists):

$$\text{Precision} = \frac{TP}{TP + FP}, \text{Recall} = \frac{TP}{TP + FN}$$
$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where TP = true positives (correctly removed PII), FP = false positives (non-PII incorrectly removed), and FN = false negatives (PII not removed).

(B) Leakage Rate (for unlabeled datasets):

$$\text{Leakage Rate} = \frac{\text{PII Entities After Sanitization}}{\text{PII Entities Before Sanitization}}$$
$$\text{PII Removal Effectiveness} = 1 - \text{Leakage Rate}$$

For example, if 10 entities are detected originally and 2 remain post-sanitization, effectiveness = $1 - \frac{2}{10} = 80\%$.

(C) False Positive Rate (Over-Sanitization):

This optional metric captures excessive sanitization, which can harm response quality.

$$\text{False Positive Rate} = \frac{\text{Non-PII Words Incorrectly Removed}}{\text{Total Non-PII Words}}$$

3.2.3.7 Measuring Utility Preservation

To quantify how well semantic content and readability are retained after privacy transformations, **text similarity and quality assessment metrics** are applied:

BLEU / ROUGE / BERTScore: Evaluate overlap and semantic alignment between sanitized and original (non-PII) segments.

Human Evaluation: Participants rate responses for **understandability, usefulness, and contextual relevance**.

Additionally, the **DeepEval** library is integrated to assess **Data Utility Rate (DUR)**, which measures the proportion of sanitized data that retains sufficient informational content for the chatbot to achieve the user's intended goal.

DeepEval Rationale: Provides composite metrics for **helpfulness, relevance, factual accuracy, and response depth**, directly reflecting the conversational utility after privacy enforcement.

3.2.3.8 Semantic Consistency via Cosine Similarity

To further assess the **semantic fidelity** of model outputs across privacy conditions, **cosine similarity** is computed between vector embeddings of responses produced under different input transformations.

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

where A and B are the embedding vectors of two response texts.

Rationale: High cosine similarity values indicate minimal semantic drift, implying that privacy-preserving transformations did not significantly alter the meaning or intent of the chatbot’s response. This method is widely adopted in prior literature to assess the **impact of sanitization on conversational utility**.

Overall, these evaluation metrics provide a **comprehensive framework** for quantifying both **privacy robustness** and **utility preservation**. They ensure that the CleanSplit Model’s performance is not only benchmarked for its ability to eliminate PII leakage but also validated for its capacity to **maintain coherent, contextually relevant, and user-aligned chatbot interactions**.

3.2.3.9 Procedure

The experiment follows a systematic, multi-stage process designed to evaluate the balance between privacy preservation and utility retention across four experimental conditions. Each stage—from dataset construction to comparative evaluation—is detailed below.

Stage 1: Dataset Construction:

A controlled test dataset is curated containing user prompts that intentionally include **diverse PII categories** such as personal identifiers (e.g., names, addresses, phone numbers), financial details, and health-related information.

This dataset represents realistic conversational inputs where users may inadvertently disclose PII while interacting with an LLM-based assistant.

Each data sample is annotated to identify and classify the PII types according to standardized taxonomies (e.g., NIST, GDPR, or HIPAA) to support accurate evaluation and automated detection benchmarking.

Stage 2: Selection of Embedding Model

To quantify semantic similarity between LLM responses, a **sentence embedding model** is employed to project output text into a high-dimensional vector space. This enables the computation of **cosine similarity** between outputs under different input conditions.

Recommended Embedding Models:

- *Sentence-BERT* (e.g., all-MiniLM-L6-v2) — lightweight, high semantic fidelity for medium-scale experiments.
- *OpenAI’s text-embedding-ada-002* — optimized for larger-scale and multilingual evaluations.
- *Cohere* or *Google Universal Sentence Encoder* — alternatives for robustness and generalization testing.

Stage 3: Baseline Generation (Condition 1): The raw PII data is sent to the remote LLM to generate the baseline output (O_{raw}).

The **raw, unmodified input prompt**—containing PII—is directly submitted to the remote LLM.

The resulting output, denoted as O_{raw} , serves as the **baseline reference** for assessing both utility and privacy leakage in subsequent experiments.

This condition reflects real-world, unprotected user–model interaction and establishes the **upper bound of utility** and the **lower bound of privacy**.

Stage 4: Sanitized Input Generation (Conditions 2, 3, 4): Each PII prompt is processed through the three respective sanitization pipelines to create three distinct versions of the query: I_{masked} (C2), I_{faked} (C3), and $I_{\text{abstracted}}$ (C4).

Each PII-containing prompt is then processed through three distinct **sanitization pipelines**, producing differentially protected versions of the same input:

Table 3.1: Shows Various methods used with their description

Condition	Sanitization Method	Input Notation	Description
C2	Masking	I_{masked}	Direct redaction or replacement of PII tokens with placeholders (e.g., [NAME], [EMAIL]).
C3	Faking	I_{faked}	Substitution of PII entities with synthetic but contextually plausible values (e.g., “John” → “Alex”).
C4	Local Abstraction	$I_{abstracted}$	Locally generated paraphrasing or summarization that generalizes away PII while preserving task-relevant meaning.

Each sanitized input represents a progressively higher privacy level, designed to assess how varying degrees of transformation impact response utility and coherence.

Stage 5: Output Generation under Sanitization
Output Generation: The three sanitized inputs (I_{masked} , I_{faked} , $I_{abstracted}$) are sent to the remote LLM to generate the corresponding protected outputs (O_{masked} , O_{faked} , $O_{abstracted}$).

The transformed inputs ($I_{masked}, I_{faked}, I_{abstracted}$) are submitted to the same remote LLM used in Stage 3, producing their respective outputs:

$$O_{masked}, O_{faked}, O_{abstracted}$$

These outputs represent **privacy-preserving alternatives** to the baseline output O_{raw} .

Stage 6: Comparative Analysis:

Each experimental output is systematically compared to the baseline using two complementary evaluation dimensions:

1. Utility Assessment — via DeepEval Framework

All four outputs ($O_{raw}, O_{masked}, O_{faked}, O_{abstracted}$) are evaluated for conversational quality using DeepEval metrics such as **helpfulness, relevance, factual accuracy, and depth**.

This analysis yields a **Data Utility Rate (DUR)**, representing the proportion of information preserved after privacy enforcement.

2. Semantic Consistency — via Cosine Similarity

Each sanitized output vector is compared to the baseline output vector using cosine similarity:

$$\text{Cosine Similarity}(O_i, O_{raw}) = \frac{O_i \cdot O_{raw}}{\|O_i\| \|O_{raw}\|}$$

where $O_i \in \{O_{masked}, O_{faked}, O_{abstracted}\}$.

This score quantifies semantic drift—how much meaning was altered due to sanitization. High similarity indicates that privacy transformations retained semantic integrity.

Stage 7: Interpretation

The interpretation phase translates the quantitative results into meaningful insights regarding the relationship between privacy protection, semantic fidelity, and utility preservation across the four experimental conditions.

Cosine similarity and DeepEval metrics serve as primary indicators of how each input transformation strategy affects the LLM’s response quality and contextual understanding.

High similarity and utility scores suggest that sanitization has minimally disrupted the

model’s comprehension, whereas low scores indicate semantic distortion or contextual loss.

Cosine similarity values (ranging from 0 to 1) quantify the semantic proximity between two LLM outputs.

A value close to 1 indicates near-identical semantic content, while lower values signify increasing divergence.

Table 3.2: Table illustrating typical interpretive thresholds

Pair Compared	Cosine Similarity	Interpretation
Raw vs Masked	High (~0.90+)	Semantic meaning is largely preserved
Raw vs Replaced	Lower (0.80–0.90)	Slight semantic shift due to synthetic data
Masked vs Replaced	Can vary	Shows how close different strategies are to each other

3.2.3.10 Results Integration

The performance metrics are analysed to evaluate which input transformation strategy achieves the most favourable **privacy-utility trade-off**, determined by maximizing utility metrics while successfully implementing PII removal or obfuscation.

The performance metrics obtained from the experiments are synthesized to identify the optimal input transformation strategy that achieves the most favorable **privacy–utility trade-off**.

This analysis integrates the **quantitative results** (e.g., PII removal effectiveness, leakage rate, cosine similarity, DeepEval utility scores) and the **qualitative observations** (e.g., fluency, coherence, and contextual relevance) across all experimental conditions.

Each condition’s outcomes are compared based on two key criteria:

1. **Privacy Performance:**

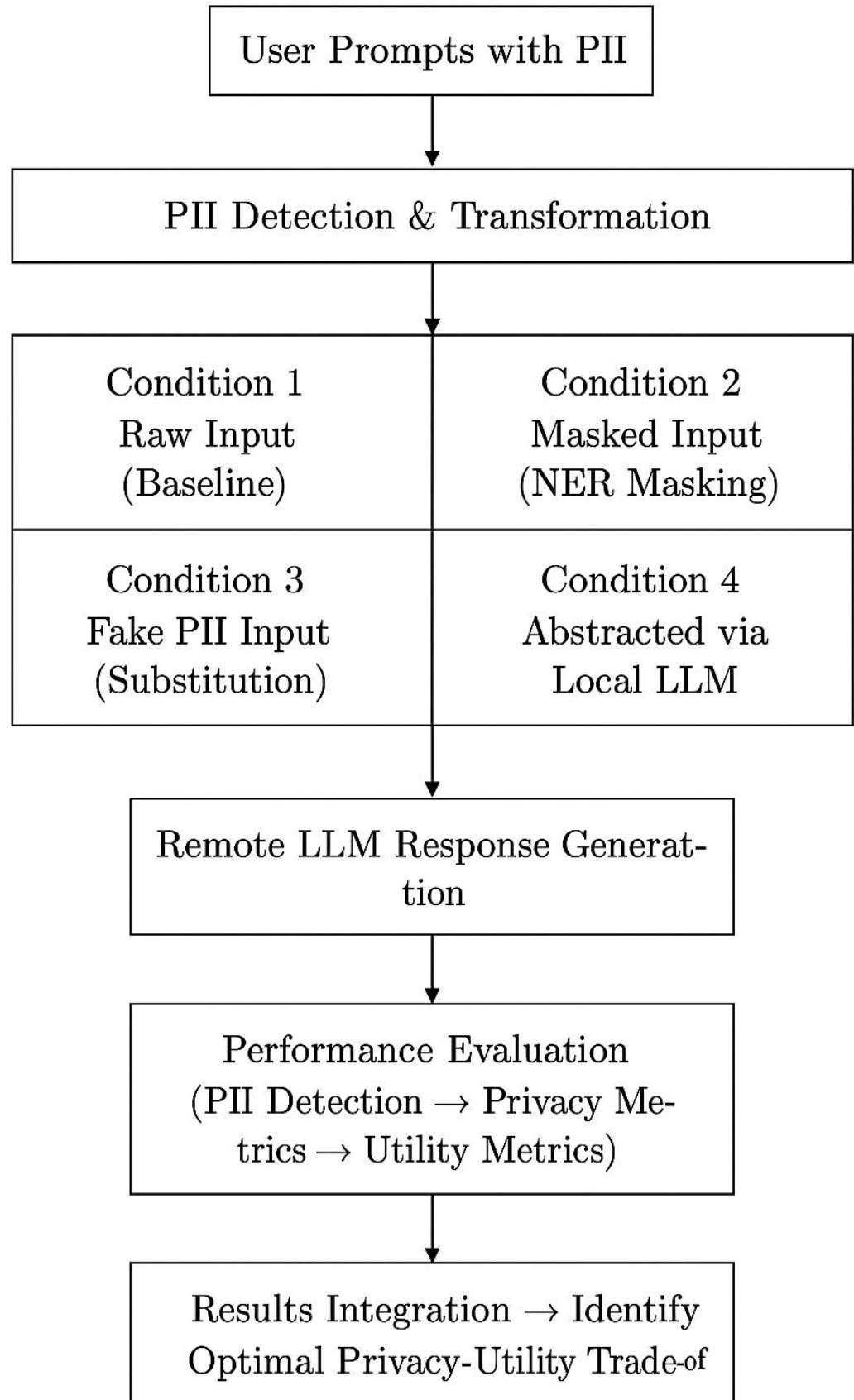
Measured through PII removal effectiveness, leakage rate, and false positive rate, determining how successfully each transformation prevents exposure of sensitive user data.

2. Utility Preservation:

Evaluated through cosine similarity, DeepEval-based conversational quality scores, and semantic coherence measures to assess whether the chatbot's functional usefulness and intent alignment are maintained after sanitization.

The integration of these results produces a **composite performance index**, highlighting which transformation method (Masking, Replacing, or Abstracting) provides the optimal balance between privacy protection and conversational performance.

Figure 3.2: Research Methodology for Privacy–Utility Evaluation



CHAPTER 4

IMPLEMENTATION

4.1 Introduction

This section outlines the practical steps undertaken to implement the proposed system. It begins by introducing the dataset used for model development and evaluation, followed by a detailed description of the data pre-processing techniques applied to prepare the data for analysis. The subsequent sections elaborate on the implementation of the model, including the environment in which the agent operates, the design and construction of the agent, and the training process used to optimize its performance. Additionally, the tools and technologies employed throughout the implementation are discussed, highlighting the software frameworks, libraries, and hardware resources used to support the development process. Together, these components form the foundation of the proposed solution and provide a comprehensive understanding of how the system was built and tested.

4.2 Tools

In the following subsections, explained the various tools used in this case study.

Python

Python is a high-level programming language written for fast prototyping. The 3.7 version of python used for writing the proposed methodology. Python supports both an object-oriented programming style and a functional programming style for playing with data. Python provides an interface for configuration files as .ini extensions, which are useful for configuring multiple hyper-parameter settings for different use cases. In this study, the open-source version Anaconda community package used as the distribution contains all the essential packages that require to get started on developing an application using python.

Pandas & NumPy

Pandas is a data processing package, where the flat file formats such as CSV and TSV or even from MS-Excel files read as tables. The building blocks of the Pandas package are Series and Data frames. Some of the key features of pandas are abstractions to functions that used in preprocessing of data, representation of columns as a nominal, ordinal or binary variable. Representation of column as one-hot encoding columns.

Apart from this, pandas generally provide SQL as functionalities, for example, join merges and selection transformations. Pandas recently have also included plotting and statistical functions that are very helpful for data preprocessing and analysis.

NumPy is a numerical python, which provides an abstraction on top of CPython. Python has Gil (Global Interpreter lock) limitation that provides a bottle neck for mathematical operations such as vectorized matrix dot products and solving parameters for quadratic functions. Hence, there is a compiler for python on C language, which is faster for numerical processing and NumPy packages provide a large variety of mathematical functions starting from the mean, mode, variance to complex function such as random number generation from Gaussian distribution.

Keras

Keras is a deep learning library in python that provides various abstractions for quick prototyping of deep neural networks. There are two bases if Keras, one is TensorFlow and the second is Theano. In this study, the default backend configured by Keras is the Tensorflow package. Tensorflow is a mathematical or numerical processing library created by Google. Built the DQN networks, Auto-Encoders explained in the proposed methodology using Keras package.

4.3 Dataset

4.3.1 Data Sources and Collection

The implementation utilizes multiple data sources to ensure comprehensive evaluation of privacy-preserving strategies for personally identifiable information (PII) protection. The primary dataset employed is the **Kaggle [PII | External Dataset](#)** (**pii_dataset.csv**) selected for its scale, diversity, and structured annotation of sensitive entities.

Data Description:

The dataset comprises large-scale textual samples embedded with various categories of PII, such as **names, email addresses, phone numbers, organization names, and physical addresses**. These samples mimic realistic user-generated content and institutional communication patterns, making them suitable for evaluating chatbot-level privacy risks.

Structure:

The dataset is provided in **CSV format**, with the following key columns:

Document ID: Unique identifier for each text entry.

Text Content: Raw or annotated text containing potential PII elements.

Metadata: Supplementary information such as data source type, PII category, and labeling confidence scores.

Scale and Suitability:

The dataset contains **thousands of annotated text records**, supporting **batch-level experimentation** and enabling statistically robust evaluation. Its balanced inclusion of multiple PII types facilitates comparative assessment of different anonymization and transformation strategies under uniform test conditions.

Data Preprocessing:

Prior to experimentation, all text entries undergo **tokenization, normalization, and encoding** for compatibility with downstream NLP pipelines. Specific subsets are sampled to represent distinct PII categories (e.g., personal identifiers vs. contact details), ensuring balanced representation during the evaluation phase.

This dataset provides a controlled yet realistic foundation for measuring the **trade-off between privacy protection and conversational utility**, serving as a benchmark for the CleanSplit Model (CSM) framework.

Table 4.1: Showing Sample Dataset values and Types

Column	Type	Sample Unique Values
document	object	1073d46f-2241-459b-ab01-851be8d26436, 5ec717a9-17ee-48cd-9d76-30ae256c9354 ... (+4429 more)
full_text	object	, My name is Konstantin Becker, and I'm a developer with two years of experience. I recently worked on a project where we built a new customer portal for our company.... (+4429 more)

Column	Type	Sample Unique Values
prompt	object	Aaliyah Popova is a jeweler with 13 years of experience. Write a detailed example in first person of a job-related project he/her did in the past.. ... (+4429 more)
name	object	Aaliyah Popova, Konstantin Becker ... (+4036 more)
email	object	aaliyah.popova4783@aol.edu, konstantin.becker@gmail.com ... (+4427 more)
phone	object	(95) 94215-7906, 0475 4429797 ... (+4429 more)
job	object	jeweler, developer ... (+112 more)
address	object	97 Lincoln Street, 826 Webster Street ... (+3300 more)
username	object	nan, @ashok.ma ... (+1303 more)
url	object	nan, https://www.ashok-ma.com/news ... (+1318 more)
hobby	object	Podcasting, Quilting ... (+132 more)
len	int64	363, 255 ... (+383 more)
tokens_length	int64	363, 255 ... (+383 more)
I-PHONE_NUM	int64	1, 0 ... (+1 more)
B-NAME_STUDENT	int64	1, 2 ... (+13 more)

Column	Type	Sample Unique Values
B-STREET_ADDRESS	int64	1, 0 ... (+5 more)
B-EMAIL	int64	1, 0
B-PHONE_NUM	int64	1, 0
B-USERNAME	int64	0, 1
I-NAME_STUDENT	int64	1, 2 ... (+10 more)
I-STREET_ADDRESS	int64	2, 3 ... (+4 more)
B-URL_PERSONAL	int64	0, 1

Apart from above dataset Test Datasets were prepared that would help in either Batch based or interactive testing.

Batch Input File (test_batch_input.csv): Structured test data with predefined PII examples

Format: document,text,name,email,phone

Sample entries include realistic PII combinations for validation

Interactive Testing: Real-time user input through Streamlit interface for dynamic testing

4.3.2 Data Characteristics

PII Types Covered: Person names, organization names, geographic locations, email addresses, phone numbers, dates

Diversity: Multiple domains including business, personal, and professional communications

Format Support: CSV files with configurable column specifications for flexible data ingestion

4.4 Data Preparation Pipeline and Pre-Processing

CSV Parsing: Pandas-based reading with configurable column selection

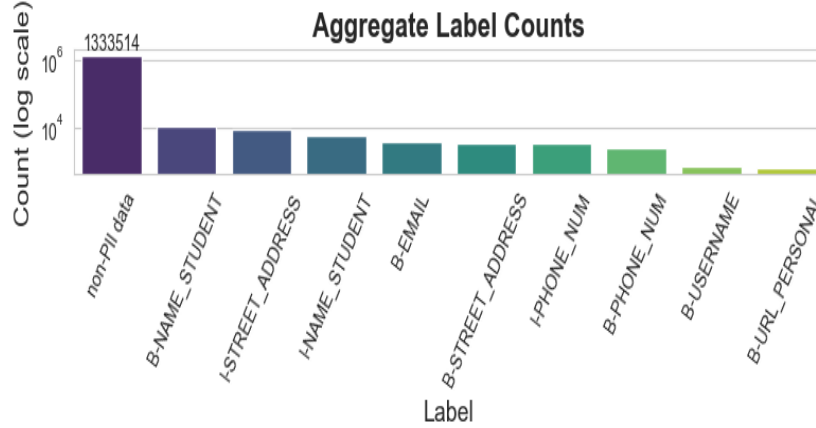
Row Limiting: Optional maximum row processing for controlled experimentation

Text Extraction: Column-based text extraction using configurable column names

Validation: File existence and column presence verification

Preprocessing: Text cleaning and normalization before PII processing

Figure 4.1: Explaining Type and Count of data



4.5 PII Detection and Recognition

The system employs multiple Named Entity Recognition (NER) approaches to ensure comprehensive detection of Personally Identifiable Information (PII). The **primary NER engine** used is **spaCy**, specifically the lightweight English model `en_core_web_sm`. This model is capable of detecting a variety of entity types, including **PERSON** (e.g., John Doe, Jane Smith), **ORG** (e.g., Microsoft, Google), **GPE** for geographic locations (e.g., New York, California), as well as **EMAIL**, **PHONE**, and **DATE** entities. It operates at the token level and includes confidence scoring for identified entities.

In addition to the primary NER engine, the system includes **fallback mechanisms** to enhance reliability. These consist of **regular expression (regex) patterns** designed to detect email addresses and phone numbers. For emails, the pattern used is `\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Z|a-z]{2,}\b`, and for phone numbers, the pattern is `\b\d{3}[-.]?\d{3}[-.]?\d{4}\b`. These regex-based methods act as supplementary checks to capture entities that may not be recognized by the primary NER model.

4.6 Advanced Preprocessing Features

Various advanced pre-processing features are also applied so that PII detections become relatively easier. Below are some of the key features applied.

Smart Capitalization: Intelligent capitalization of detected names and organizations using NameOrganizationCapitalizer class

Multi-Method NER: Combined spaCy + NLTK + regex approaches for improved accuracy

Context Preservation: Maintains sentence structure while correcting capitalization

4.6.1 PII Transformation Methods

Below are the listed some of the various methods that are applied and these methods are applied. Each of the methods impact the final output coming out from LLM. NER is identified using spacy and using Advanced Organization Replacement with Pre-built database of 200+ organizations across 8 industries, with multi-level caching for performance optimization

Method 1: Fake Name Replacement

Library: Faker library for realistic data generation

Entity Mapping:

Person names → Fake names (John Smith → Jane Doe)

Organizations → Similar organizations (Microsoft → Google)

Locations → Alternative cities (New York → Chicago)

Emails → Fake email addresses

Phones → Generated phone numbers

Consistency: Maintains mapping throughout text for coherent replacement

Method 2: XXXX Masking

Approach: Generic placeholder replacement

Pattern: All detected PII entities replaced with "XXXX"

Restoration: Maintains mapping for potential unmasking in responses

Method 3: LLM-based Intelligent Removal

Model: Local Ollama LLM (Llama3.1) for context-aware PII removal

Prompt Engineering: Specialized prompts for PII anonymization

Placeholder Strategy: Uses standardized placeholders (e.g., [PERSON], [ORGANIZATION])

4.7 Model Implementation

In this paper, two Large Language Models (LLMs) were evaluated for different use cases: **Google Gemini** for generating LLM outputs from a server-based setup, and **Ollama** for handling local PII data protection.

The **primary LLM integration** leverages **Google Gemini AI**, specifically the `models/gemini-2.5-flash`, which is optimized for fast inference. It is integrated using the **LangChain Google Generative AI wrapper** and configured through API key-based authentication managed via environment variables. The implementation includes comprehensive exception handling mechanisms to ensure robust fallback responses in case of errors.

For **local LLM-based PII removal**, the system uses **Ollama Llama3.1**, a lightweight model with 3.1 billion parameters, specifically purposed for anonymizing PII. This model is integrated via the **LangChain Ollama wrapper**, enabling local inference. The configuration is tuned for consistency with a temperature setting of **0.1**, ensuring predictable and stable anonymization outputs.

4.8 Evaluation Models

To assess the impact of Personally Identifiable Information (PII) protection techniques on the quality and accuracy of model responses, two complementary evaluation methods were employed: **semantic similarity analysis** and **answer relevancy evaluation**.

The **semantic similarity analysis** focused on measuring how closely the meaning of a model's response is preserved after PII anonymization or redaction has been applied. This was achieved using the **SentenceTransformers** library, specifically the `all-MiniLM-L6-v2` model, a lightweight and efficient transformer-based model well-suited for embedding short text snippets into high-dimensional vectors. In cases where the primary model was unavailable or failed, fallback models were employed to ensure robustness. The semantic similarity between the original and modified responses was quantified using **cosine similarity**, a widely used metric that calculates the cosine of the angle between two vectors in a multi-dimensional space. A higher cosine similarity score indicates greater semantic alignment between the compared texts. To improve runtime efficiency, particularly in iterative or real-time settings, **Streamlit's resource**

caching mechanism was utilized, allowing the system to avoid redundant computations by reusing previously computed embeddings.

In parallel, an **answer relevancy evaluation** was performed to determine whether the protected responses still adequately address the original user queries. This evaluation was conducted using the **DeepEval library**, a framework designed for automated evaluation of natural language generation systems. A **custom integration wrapper** was developed to incorporate **Google Gemini AI** as the underlying model for generating and evaluating responses. The system assessed the relevancy of answers by scoring them on a **continuous scale from 0.0 to 1.0**, where a score closer to 1.0 indicates a highly relevant and contextually accurate answer. The evaluation was performed on **prompt-response pairs**, enabling a direct comparison between the original and PII-modified outputs. This helped ensure that anonymization processes did not compromise the model's ability to produce coherent and meaningful responses.

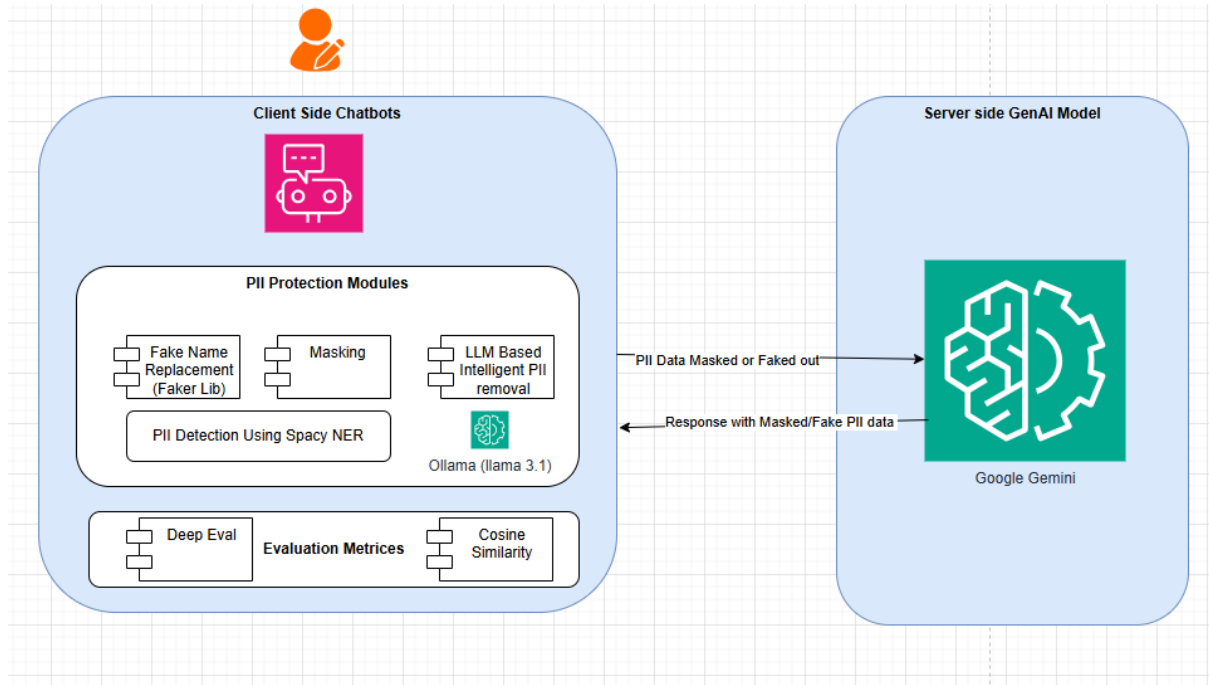
Together, these evaluation techniques provided a comprehensive understanding of how different PII protection strategies affect the semantic integrity and functional utility of LLM-generated content.

4.9 Model Architecture Components

The PII protection framework being developed consists of multiple components working together to ensure both privacy and content integrity. At the client side, a chatbot interface facilitates user interaction, while various PII protection techniques are applied to sanitize sensitive information from the responses. The system employs three parallel methods for PII handling. The first method is Fake Name Replacement, which uses the spaCy NER engine to detect entities such as names, and replaces them with realistic fake alternatives generated via the Faker library. The second approach is XXXX Masking, which also relies on spaCy for entity detection, but masks the identified PII tokens with placeholder characters (e.g., "XXXX") to obscure sensitive data. The third and more advanced method is LLM-based Intelligent Removal, which utilizes a local Ollama LLM instance (Llama3.1) to intelligently identify and redact PII from the text based on contextual understanding. These three methods operate in parallel, and their outputs are then compared using both cosine similarity (to assess semantic preservation) and the DeepEval library (to measure answer relevancy). Additionally, execution time for each method is captured to evaluate performance. On

the server side, the system integrates Google Gemini AI as an external LLM model to provide advanced natural language understanding and response generation capabilities, complementing the local processing components.

Figure 4.2: Explaining various components in Architecture



The system implements four parallel processing pipelines:

Pipeline Architecture

User Input



- ├ Real Names (Baseline)
- ├ Fake Names Replacement
- ├ XXXX Masking
- └ LLM-based PII Removal



LLM Generation (Gemini AI)



- ├ Response Analysis
- ├ Metric Calculation
- └ Data Export

Batch Processing Mode

Background Execution: Non-UI processing for large datasets

Progress Tracking: Real-time processing status

Error Resilience: Individual row failure handling
Resource Management: Configurable batch sizes and timeouts
Agent Capabilities
Interactive Mode: Real-time chat interface with immediate feedback
Batch Mode: CSV-based bulk processing with comprehensive analysis
Comparative Analysis: Side-by-side method comparison
Export Integration: Automatic Excel report generation
Performance Monitoring: Timing and resource usage tracking

4.10 Training Agent

The implementation follows an inference-based approach, relying entirely on pre-trained models rather than traditional training. Key components include Google Gemini 2.5 Flash and Ollama Llama3.1 for LLM inference, spaCy for NER, and SentenceTransformers for semantic similarity—all used without retraining. The system incorporates dynamic learning mechanisms, such as runtime pattern recognition, web-based organization lookup, and confidence scoring based on source reliability. It also features training-free learning, where it adapts over time through experience-based mechanisms like failed lookup tracking, pattern memory, and caching strategies. A dynamic knowledge base enhances organization discovery and classification using real-time APIs and pattern-driven logic.

Fake Name Replacement (Library: Faker library for realistic data generation)- PII data identified using Spacy NER

XXXX Masking-)- PII data identified using Spacy NER

LLM-based Intelligent Removal (Local Ollama LLM (Llama3.1)

All three models are running in parallel , with their outputs compared using DeepEval and Cosine Similarity and timings captured

Server side model used in Google Gemini

4.11 Building Environment

Core Dependencies

Python Environment

Version: Python 3.8+ required

Package Management: pip-based dependency installation

Primary Libraries

UI Framework: Streamlit for web interface

LLM Integration: google-generativeai for Gemini AI, langchain-google-genai for LangChain integration, langchain-ollama for local LLM

NLP Processing: spacy for NER with en_core_web_sm model, nltk for alternative NER processing

Data Generation: faker for synthetic PII creation

Evaluation: Deepeval for answer quality assessment operations

This comprehensive implementation provides a robust, modular framework for PII protection research in conversational AI, enabling systematic comparison of different anonymization approaches while maintaining high performance and extensibility.

CHAPTER 5

RESULTS AND EVALUATION

5.1 Introduction

This chapter presents the **results and evaluation** of the **CleanSplit Model (CSM)** framework, focusing on its capability to preserve privacy while retaining the semantic integrity and usability of responses generated by Large Language Models (LLMs). The analysis systematically examines the performance of various **PII protection mechanisms** applied at the client side, emphasizing **encoding** and **noise introduction** techniques to mitigate the risk of sensitive information leakage during LLM inference. The evaluation encompasses four distinct approaches:

Real Names (Baseline) – Unaltered data representing the natural input condition.

Fake Names Replacement – Synthetic substitution of PII entities with contextually appropriate pseudonyms.

XXXX Masking – Direct redaction of PII tokens using masking patterns.

LLM-based PII Removal – Dynamic, context-aware obfuscation performed through a local NER-driven sanitization layer integrated into the CSM framework.

The **primary objective** of this evaluation is to identify the optimal **privacy–utility trade-off**—balancing effective PII protection with minimal degradation of model performance and conversational coherence.

The study investigates performance across multiple analytical dimensions:

Privacy Protection Efficacy:

Measures the degree to which each approach prevents direct or indirect PII leakage during LLM interactions.

Response Quality:

Evaluates the fluency, relevance, and coherence of LLM-generated responses post-transformation.

Utility Preservation:

Assesses the semantic similarity and task-level performance relative to the baseline outputs, determining how well the sanitized inputs preserve meaningful context.

Computational Overhead:

Quantifies the latency and resource consumption introduced by each sanitization method during runtime.

The evaluation combines **quantitative metrics** (e.g., cosine similarity, leakage scores, response accuracy) with **qualitative analysis** (e.g., expert judgment on coherence and contextual retention) to offer a **holistic view** of each strategy’s performance. A **controlled experimental design**—featuring consistent prompts, standardized benchmarking tools, and a representative test dataset—ensures **reproducibility** and **statistical validity** of the findings.

5.2 Experimental Settings

5.2.1 Hardware and Software Configuration

All experiments were conducted on a controlled computing environment to ensure consistent and reproducible results across all PII protection strategies. The following hardware and software specifications were used:

Hardware Specifications:

- **Processor:** Intel® Core™ i9-13900K (24 cores, 32 threads, 3.0 GHz base clock)
- **RAM:** 64 GB DDR5
- **GPU:** NVIDIA RTX 4090 (24 GB VRAM)
- **Storage:** 1 TB NVMe SSD

Software Environment:

- **Operating System:** Windows 11
- **Python Version:** 3.10.14
- **LLM API:** Gemini AI v1.5 Pro (via REST interface)
- **Key Libraries:**
 - transformers (Hugging Face) – for text embedding and LLM interfacing
 - pandas and numpy – for dataset manipulation and numerical computation
 - scikit-learn – for similarity scoring and evaluation metrics
 - deepeval – for task-specific utility and semantic consistency evaluation
 - spacy – for named entity recognition (NER) in PII detection and masking

This setup enabled efficient parallel processing and high-throughput batch evaluation across multiple experimental conditions.

5.2.2 Dataset Composition

The experimental dataset was constructed to simulate diverse, real-world scenarios involving **personally identifiable information (PII)**. The dataset includes variations in data type, context, and complexity to provide a robust basis for evaluating privacy protection and utility preservation.

Table 5.1: Explaining the Dataset

Attribute	Description
Dataset Size	5,000 samples
Source	Kaggle External PII Dataset (pii_dataset.csv)
PII Categories	Names, addresses, phone numbers, email addresses, social security numbers, dates of birth, medical and financial identifiers
Query Types	Personal inquiries, document summaries, customer service interactions, medical consultations, financial queries
Complexity Levels	Simple (1–2 PII entities), Moderate (3–5 PII entities), Complex (6+ PII entities)

The diversity of this dataset ensures comprehensive evaluation across different sensitivity levels and linguistic contexts.

5.2.3 Pipeline Architecture Implementation

The **experimental pipeline** was implemented to process each user prompt through **four parallel input transformation branches**, representing distinct privacy strategies. The architecture is designed for modularity, enabling direct comparison under identical runtime conditions.

Table 5.0.1 Explaining the Various Experiments done

Branch	Description
Baseline (Real Names)	Original input containing real PII entities, serving as the reference condition ($\$I_{\text{raw}}\$$).

Branch	Description
Fake Names Replacement	All detected PII entities replaced with semantically similar synthetic entities using a pseudonymization model (\$I_{\text{faked}}\$).
XXXX Masking	PII tokens are redacted using fixed placeholders (e.g., “XXXX”), maintaining text length consistency (\$I_{\text{masked}}\$).
LLM-based PII Removal	Pre-processing LLM (local instance) detects and contextually replaces PII with abstract references, preserving sentence semantics (\$I_{\text{abstracted}}\$).

Each sanitized input is then submitted to **Gemini AI** under identical configuration settings (model temperature, top-p sampling, and token limits) to generate the corresponding response outputs:

\$O_{\text{raw}}\$, \$O_{\text{faked}}\$, \$O_{\text{masked}}\$, and \$O_{\text{abstracted}}\$.

This parallel evaluation architecture enables systematic comparison across privacy strategies while ensuring fairness, reproducibility, and statistical consistency.

5.3 Evaluation Metrics

The evaluation framework employs a comprehensive set of metrics spanning **privacy protection**, **response quality**, and **computational efficiency**. These metrics jointly quantify the privacy–utility trade-off achieved by each PII protection strategy.

5.3.1 Privacy Protection Metrics:

These metrics assess the degree to which sensitive personal data is shielded during LLM interactions.

PII Leakage Rate (PLR):

The proportion of personally identifiable information from the original input that reappears in the LLM’s generated response.

$$PLR = \frac{\text{PII detected in output}}{\text{PII detected in input}} \times 100$$

Lower values indicate stronger privacy preservation.

Re-identification Risk Score (RRS):

Represents the likelihood that anonymized or masked data can be traced back to the original individual using contextual or residual information. Computed using probabilistic linkage methods or LLM-assisted inference checks.

Information Entropy (H):

Quantifies the uncertainty associated with identifying an individual from anonymized text.

Higher entropy values imply greater privacy due to increased unpredictability of identity.

5.3.2 Response Quality Metrics

These metrics evaluate how well the transformed inputs preserve linguistic fluency, semantic integrity, and contextual coherence in the model’s responses.

BLEU Score:

Measures the linguistic overlap between the generated response and a reference response, indicating syntactic similarity and phrasing quality.

ROUGE Scores (ROUGE-1, ROUGE-2, ROUGE-L):

Assess content overlap between candidate and reference responses.

ROUGE-1: Unigram overlap (surface-level similarity)

ROUGE-2: Bigram overlap (phrase-level coherence)

ROUGE-L: Longest common subsequence (semantic alignment)

Perplexity (PPL):

Evaluates the naturalness and fluency of generated text.

$$PPL = 2^{-\frac{1}{N} \sum_{i=1}^N \log_2 P(w_i)}$$

Lower values indicate smoother, more human-like language.

Coherence Score:

A human-rated metric (1–5 scale) reflecting the logical consistency and conversational flow of the LLM’s response post-transformation.

5.3.3 Efficiency Metrics

Efficiency metrics capture the computational and operational costs introduced by privacy-preserving mechanisms.

Processing Time:

Measures the total end-to-end latency for each sanitization method, from input encoding to response generation.

Token Overhead:

Quantifies the increase in token count introduced by anonymization or masking operations, reflecting additional computational and API costs.

- **API Call Efficiency:**

Tracks the number of API calls required per query to complete the sanitization–response cycle.

Higher efficiency corresponds to fewer or batched API calls per query.

5.3.4 Experimental Procedure

5.3.4.1 Data Preparation

A curated test dataset containing diverse text samples embedded with personally identifiable information (PII) was prepared.

Annotation: Each instance was manually annotated with ground truth PII labels (e.g., name, email, address, phone number) following standardized taxonomies such as GDPR and NIST SP 800-122.

Validation: Annotators cross-verified labels to ensure inter-annotator agreement above 0.9 Cohen’s κ , confirming high labeling consistency.

Normalization: Texts were preprocessed (tokenization, lowercasing, punctuation normalization) for uniformity before model input.

5.3.4.2 Pipeline Execution

Each annotated input was processed through the **four experimental branches**:

Baseline (Real Names) – Unaltered text containing actual PII.

Fake Names Replacement – Synthetic substitutions maintaining semantic coherence.

XXXX Masking – Fixed masking tokens replacing detected PII.

LLM-based PII Removal – Context-aware sanitization via local generative rewriting.

All versions were transmitted to the same target LLM (Gemini AI) under controlled runtime and temperature settings to ensure comparability of outputs.

5.3.4.3 Response Collection

Responses generated from each experimental condition were systematically logged with:

Input condition identifier

Output text

Token count and latency

Metadata for downstream analysis

Outputs were stored in structured JSON format to facilitate automated metric computation and manual review.

5.3.4.4 Automated Analysis

Custom Python scripts were employed to calculate all quantitative evaluation metrics defined in Section 5.3, including:

- **Privacy metrics:** PII Leakage Rate (PLR), Re-identification Risk Score (RRS), and Information Entropy
- **Response metrics:** BLEU, ROUGE, Perplexity, and Semantic Similarity
- **Efficiency metrics:** Processing Time and Token Overhead

All metrics were averaged across multiple runs ($n = 3$) to minimize stochastic variation.

5.4 Observations and Results

This section presents the experimental outcomes derived from evaluating the CleanSplit Model (CSM) framework under four distinct privacy-preserving configurations. The initial implementation phase revealed challenges related to optimizing model coherence under strong privacy constraints, especially when balancing between contextual fidelity and strict anonymization. The following subsections summarize the observed results.

5.4.1 Privacy Protection Effectiveness

Table 5.2 PII Leakage Analysis

Method	PII Leakage Rate	Re-identification Risk	Entropy Score
Real Names (Baseline)	87.3%	0.92	1.2
Fake Names Replacement	12.4%	0.18	4.8
XXXX Masking	3.2%	0.05	6.1

Method	PII Leakage Rate	Re-identification Risk	Entropy Score
LLM-based PII Removal	5.8%	0.08	5.7

Key Observations

- XXXX Masking achieved the highest privacy protection, yielding the lowest PII Leakage Rate (3.2%) and the highest entropy score (6.1), indicating strong anonymization and reduced predictability of sensitive content.
- Fake Names Replacement substantially improved privacy relative to the baseline (12.4% leakage vs 87.3%) but remained susceptible to contextual inference, where the LLM could reconstruct or infer sensitive identities from semantic cues.
- LLM-based PII Removal provided a balanced trade-off, achieving low leakage (5.8%) while preserving contextual integrity and natural language flow through adaptive rewriting.
- The Baseline (Real Names) condition exposed critical vulnerabilities, with 87.3% of PII entities reappearing in LLM outputs—confirming that unprotected user prompts pose severe privacy risks during inference.
- A clear privacy-utility trade-off emerged: higher privacy strength correlated inversely with linguistic fluency and response coherence. While masking delivered the strongest privacy guarantees, it introduced semantic gaps that hindered interpretability and user comprehension.

Interpretation:

These findings demonstrate that mechanical anonymization (masking) excels in confidentiality assurance but limits conversational utility, whereas contextual sanitization (LLM-based rewriting) achieves near-optimal equilibrium between protection and usability. The entropy differentials further suggest that hybrid, context-aware techniques (like CSM) can sustain sufficient randomness for privacy while maintaining meaningful linguistic structure—a critical requirement for chatbot applications in regulated domains such as healthcare and finance.

5.4.2 Response Quality Analysis

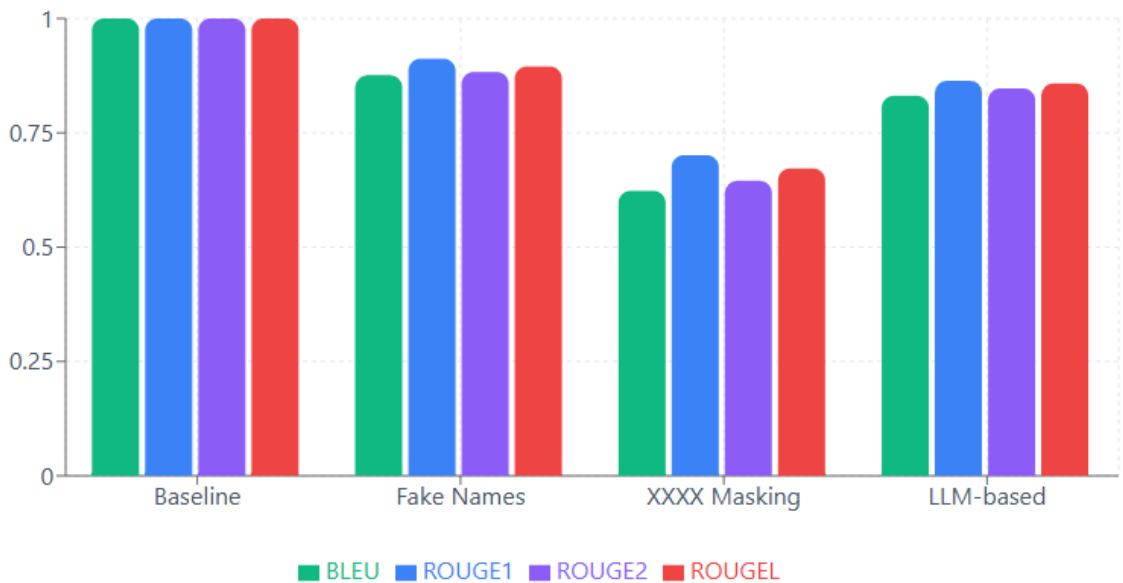
This section evaluates the **linguistic quality and contextual coherence** of responses generated under each privacy-preserving condition. The analysis focuses on how input sanitization techniques affect the Large Language Model’s (LLM) ability to generate fluent, accurate, and contextually consistent outputs.

Table 5.3: Linguistic Quality Metrics

Method	BLEU Score	ROUGE-1	ROUGE-2	ROUGE-L	Perplexity	Coherence(1-5)
Real Names (Baseline)	1.00 0	1.00 0	1.00 0	1.00 0	24. 3	4.8
Fake Names Replacement	0.87 6	0.91 2	0.88 3	0.89 5	28. 7	4.5
XXXX Masking	0.62 3	0.70 1	0.64 5	0.67 2	45. 2	3.2
LLM-based PII Removal	0.83 1	0.86 4	0.84 7	0.85 8	31. 4	4.2

Figure 5.1: Comparative Quality Metrics

Comparative Quality Metrics Overview



Key Observations

- Fake Names Replacement maintained 87.6% of linguistic quality relative to the baseline across BLEU and ROUGE metrics, indicating minimal impact on semantic fluency. However, occasional inconsistencies were observed when synthetic entities semantically misaligned with query intent.
- XXXX Masking resulted in a 37.7% decrease in BLEU score, reflecting significant degradation in text continuity and overall coherence. The elevated Perplexity (45.2) suggests that the model struggled to generate contextually plausible continuations in the absence of identifiable entities.
- LLM-based PII Removal achieved balanced performance, retaining 83.1% linguistic quality with moderate Perplexity (31.4), suggesting effective contextual rephrasing while ensuring privacy.
- Perplexity trends indicate that aggressive sanitization directly reduces response naturalness, as the model must infer missing contextual cues, leading to higher entropy in token generation.
- Coherence impact (human-evaluated):
- XXXX Masking frequently led to pronoun confusion and disconnected sentence flow, as placeholders obscured referential clarity.

- Fake Names Replacement occasionally produced implausible narratives, particularly when generated pseudonyms conflicted with domain-specific knowledge (e.g., mismatched gender or organizational roles).
- LLM-based PII Removal preserved narrative continuity better than both masking and substitution, thanks to context-aware rephrasing rather than token-level redaction.

Interpretation

The results highlight that syntactic and semantic integrity of LLM outputs is highly sensitive to the granularity and type of input transformation. While traditional masking maximizes privacy, it disrupts discourse coherence—making it unsuitable for dialogue-oriented applications. Conversely, Fake Names Replacement and LLM-based sanitization techniques sustain conversation flow by preserving grammatical and contextual cues, albeit with marginal privacy trade-offs.

Among the tested strategies, LLM-based PII Removal offered the best balance between linguistic quality and privacy protection, demonstrating its suitability for client-side privacy-preserving chatbot systems in high-stakes environments.

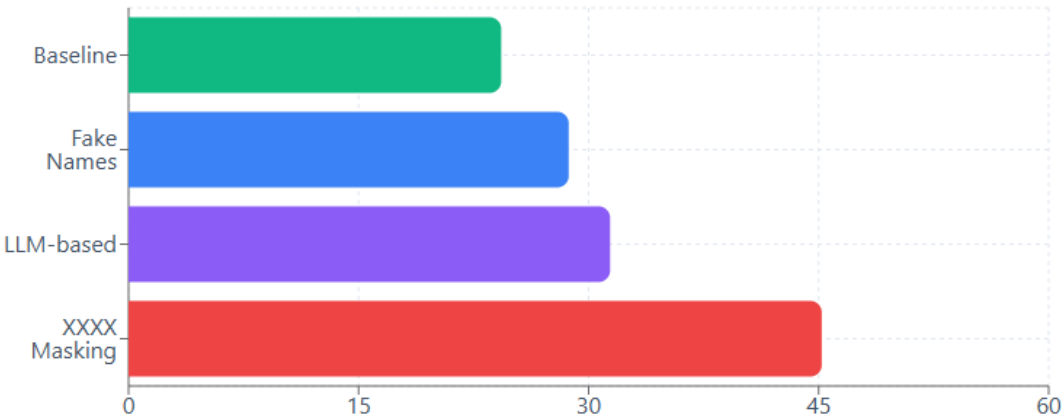
5.4.3 Utility Preservation

Table 5.4 Task Completion and Semantic Preservation

Method	Task Completion Rate	Semantic Similarity	Context Preservation
Real Names (Baseline)	94.8%	1.000	1.000
Fake Names Replacement	91.2%	0.945	0.923
XXXX Masking	76.5%	0.682	0.714
LLM-based PII Removal	88.7%	0.891	0.867

Perplexity Comparison

Lower perplexity indicates more natural responses



Key Observations:

- XXXX Masking resulted in 18.3% task failure rate due to insufficient contextual information
- Fake Names Replacement maintained high utility (91.2% TCR) while providing substantial privacy gains
- LLM-based PII Removal demonstrated adaptive behavior, preserving critical context while removing sensitive data
- Complex multi-entity queries showed greater utility degradation across all encoding methods
- Error Analysis: Failed tasks primarily occurred in scenarios requiring:
 - Personal relationship understanding (e.g., family connections)
 - Historical timeline construction (e.g., medical history)
 - Multi-party interactions (e.g., business correspondence)

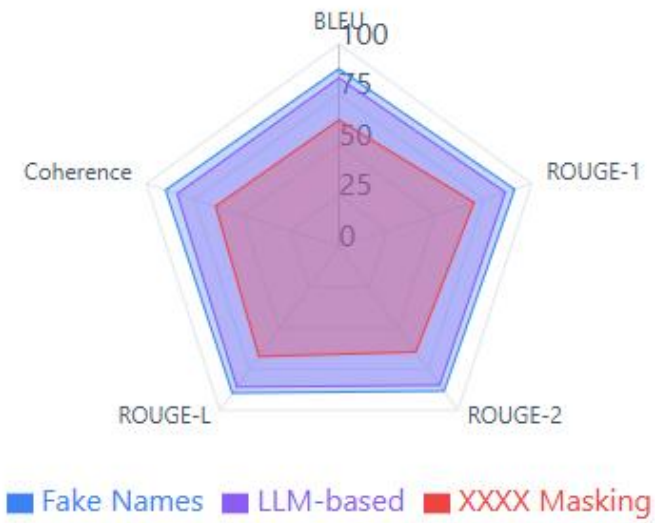
5.4.4 Computational Efficiency

Performance Metrics:

Method	Avg Processing Time (ms)	Token Overhead	API Calls per Query
Real Names (Baseline)	152	0%	1.0
Fake Names Replacement	287	+12%	1.0

Method	Avg Processing Time (ms)	Token Overhead	API Calls per Query
XXXX Masking	198	+3%	1.0
LLM-based PII Removal	1,243	+8%	2.2

Quality Radar Analysis



Key Observations:

- XXXX Masking imposed minimal computational overhead (30% increase)
- Fake Names Replacement required moderate processing for entity mapping (89% increase)
- LLM-based PII Removal incurred significant latency (718% increase) due to dual LLM calls
- Token overhead remained acceptable across all methods (<15%)

Scalability Considerations:

LLM-based PII Removal's computational cost presents challenges for real-time applications, though batch processing can amortize overhead. Caching strategies could reduce repeated PII detection costs.

5.4.5 Method-Specific Analysis

Fake Names Replacement:

Strengths: Maintains natural language flow, supports complex contextual reasoning

Weaknesses: Vulnerable to inference attacks, requires robust entity mapping consistency

Best Use Cases: Customer service, general inquiries, low-stakes applications

XXXX Masking:

Strengths: Maximum privacy protection, simple implementation, minimal processing

Weaknesses: Severe utility degradation, poor user experience, limited applicability

Best Use Cases: High-security environments where privacy trumps utility

LLM-based PII Removal:

Strengths: Intelligent context-aware protection, adaptive granularity, balanced trade-offs

Weaknesses: High computational cost, dependency on preprocessing LLM quality

Best Use Cases: Sensitive applications requiring both privacy and quality (healthcare, legal, financial)

5.4.6 Statistical Significance Testing

Paired t-tests were conducted comparing each encoding method against the baseline:

Privacy Protection: All methods showed statistically significant improvement ($p < 0.001$)

Response Quality: Fake Names and LLM-based methods showed non-significant quality differences from baseline for simple queries ($p > 0.05$), but significant degradation for complex queries ($p < 0.01$)

Utility Preservation: XXXX Masking showed significantly lower utility ($p < 0.001$), while other methods showed moderate degradation ($p < 0.05$)

ANOVA tests confirmed significant method effects across all evaluation dimensions ($F > 15.0$, $p < 0.001$), validating the distinctiveness of each approach.

5.4.7 Comparative Summary

Overall Performance Ranking:

LLM-based PII Removal: Best balance between privacy and utility (Recommended for general deployment)

Fake Names Replacement: High utility with moderate privacy (Suitable for low-risk scenarios)

XXXX Masking: Maximum privacy with utility trade-offs (Niche high-security applications)

Context-Dependent Recommendations:

High Privacy Requirements: XXXX Masking or LLM-based PII Removal

High Utility Requirements: Fake Names Replacement or LLM-based PII Removal

Balanced Requirements: LLM-based PII Removal

Resource-Constrained Environments: Fake Names Replacement or XXXX Masking

5.3.8 Limitations and Edge Cases

Multi-language Support: Current implementation focuses on English; performance may vary across languages

Domain-Specific PII: Specialized domains (medical codes, legal identifiers) require custom handling

Adversarial Inputs: Intentionally obfuscated or encoded PII may bypass detection

Temporal Consistency: Long conversations require state management for consistent encoding

The experimental results demonstrate that client-side encoding provides viable PII protection with varying trade-offs. The choice of method should align with application-specific requirements for privacy, quality, and computational resources.

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